Relativity and Cosmology II Lecture Note

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June 08, 2026

Abstract

This is the lecture note of the GR 2 course in EPFL 2026 spring semester.

Contents

1 Lecture 1: FRW Metric

For a cosmological model, we need to make some assumptions about the universe. The most common assumption is the cosmological principle:

Theorem 1.1 **Cosmological Principle**

The universe is homogeneous and isotropic on large scales.

The cosmological principle strongly constains the form of the metric. We assume that we can select a space slice at each time in some coordinate system, and consider the induced metric on each time slice.

1.1 Space Slices Metric

The cosmological principle requires that the metric on each time slice is homogeneous and isotropic, which means that the metric on each time slice is a maximally symmetric space. A maximally symmetric space is a space that has the maximum number of symmetries, which means that the Riemann curvature tensor has the following form:

$$\begin{aligned} &R^{(3)} = \text{constant} \\ &R_{ij} = \frac{1}{3} R^{(3)} g_{ij} \\ &R^{(3)}_{ijkl} = \frac{1}{6} R^{(3)} \left(g_{ik} g_{jl} - g_{il} g_{jk} \right) \end{aligned} \tag{1}$$

Thus we can see that a spactial metric can be classified into three types, and in fact each type of metric can be written in a simple form:

- $R^{(3)} = 0$, the metric is $\mathbb{R}^3$ 3-plane
- $R^{(3)} > 0$, the metric is $\mathbb{S}^3$ 3-sphere
- $R^{(3)} < 0$, the metric is $\mathbb{H}^3$ 3-hyperbolic space (Euclidean AdS space)

The metirc of $\mathbb{R}^3$ is trivial, we will consider the other two cases in the following:

1.1.1 $\mathbb{S}^3$ 3-sphere

The metric of a 3-sphere can be given by inducing from a 4-dimensional Euclidean space with Cartesian coordinates x^i and the metric $ds^2 = \delta_{ij} dx^i dx^j$. We can induce a coordinate and metric by the following constaint and parametrization:

- Constraint: $\sum_i x_i^2 = a^2$
- Parametrization:

$$\begin{aligned} x_1 &= R \sin\theta \cos\varphi \\ x_2 &= R \sin\theta \sin\varphi \\ x_3 &= R \cos\theta \\ x_4 &= \sqrt{a^2 - R^2} \end{aligned} \tag{2}$$

The induced metric is given by:

$$ds^2 = \frac{a^2 dr^2}{a^2 - r^2} + r^2 d\Omega_2^2 \tag{3}$$

We can calculate the Ricci Scalar of this metric, and we find that $R^{(3)} \sim \frac{1}{a^2}$.

1.1.2 $\mathbb{H}^3$ 3-hyperbolic space

Similarly we can embed the 3-hyperbolic space into a 4-dimensional Minkowski space with Cartesian coordinates x^i to get the metric. Another easy way of seeing the result is simply take the metric of the 3-sphere and replace a^2 with $-a^2$. This will definitely give us a solution to the constraints Equation 1 with negative curvature. Thus we can write the metric of the 3-hyperbolic space as:

$$ds^2 = \frac{a^2 dr^2}{a^2 + r^2} + r^2 d\Omega_2^2 \tag{4}$$

1.1.3 Unified Form of Spacial Metric

With above preparations, we can write three types of metrics in a unified form:

$$ds^2 = a^2 \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega_2^2 \right) \tag{5}$$

Here $k = -1, 0, 1$ stands for the three types of metrics. $k = 1$ gives $\mathbb{S}^3$ and $k = -1$ gives $\mathbb{H}^3$.

1.2 FRW Metric

1.2.1 Definition in Comoving Coordinates

With the spacial slice metric prepared, we can add a time coordinate to make it a full 4 dimensional metric ansatz. A typical form of the metric is given by:

Definition 1.2 **FRW Metric**

The FRW metric is given by:

$$\begin{aligned} ds^2 &= -dt^2 + a^2(t) \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega_2^2 \right) \\ &= -dt^2 + a^2(t) \gamma_{ij} dx^i dx^j \end{aligned} \tag{6}$$

Mathematically, we can rigorously show that the FRW metric is the only metric that satisfies the cosmological principle. Physically, we can also understand that the FRW metric is the most general metric that can describe a homogeneous and isotropic universe.

Remark

However, we shall notice that such metric is only defined on a patch of the manifold. Different manifold with different topology can have the same local metric. Thus, the global structure of the spacetime is undetermined.

1.2.2 Unit of the Scale Factor

We can see that:

- For $k \neq 0$:, the scale factor $a(t)$ has the physical meaning of radious of spacial curvature. Thus, we can assign it a unit of length and take the spacial coodinate dimesionless.
- For $k = 0$: the scale factor $a(t)$ can be taken as dimensionless, and the spacial coordinate can be taken as having a unit of length.

In fact in this case, at fixed time, $a(t = t_0)$ is meaningless, what physical is the ratio $\frac{a(t)}{a(t_0)}$.

1.2.3 Remark on Comoving Coordinates

Definition 1.2 is certainly written in a certain coordinate system, which is called the **comoving coordinates**. This means that the particles at rest in this frame (at rest means at rest spacially and moving in the time coordinate) are free (which means they move in geodesics) We can see that indeed $u^0 = 1$ is a solution to the geodesic equation.

1.2.4 Connection and Curvature

We can calculate the Levi-Civita connection and the curvature of the FRW metric in the comoving coordinates:

- Metric:

$$g_{00} = -1, \quad g_{ij} = a^2\gamma_{ij} \quad g^{00} = -1, \quad g^{ij} = a^{-2}\gamma^{ij} \tag{7}$$

- Curvature:

$$\begin{aligned} R_{00} &= -3\frac{\ddot{a}}{a} \quad R_{0i} = 0 \\ R_{ij} &= (a\ddot{a} + 2\dot{a}^2 + 2k)\gamma_{ij} \\ R &= 6\frac{a\ddot{a} + \dot{a}^2 + k}{a^2} \end{aligned} \tag{8}$$

where γ_{ij} is the spacial metric. We can see that the curvature is determined by the scale factor $a(t)$ and the curvature parameter k.

2 Lecture 2: Friedmann Equations and Solutions

We can rewrite the Einstein Field Equations with two ansatz:

- The metirc is given by the Friedmann-Robertson-Walker (FRW) metric
- The EM tensor of matter is given by a perfect fluid in the comoving frame

2.1 Einstein Field Equations

We consider the general Einstein Field Equation with a cosmological constant:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \tag{9}$$

2.1.1 Matter Energy-Momentum Tensor in Comoving Frame

In Section 1.2.4 we have calculated the curvature tensor in the comoving frame, which is the LHS of Equation 9. Now we need to calculate the EM tensor of matter in the comoving frame. We focus on:

- The matter is a perfect fluid and being at rest in the comoving frame

which means that the EM tensor has the following form in the comoving frame:

$$T_{\mu\nu} = u_\mu u_\nu (\rho + p) + p g_{\mu\nu} \tag{10}$$

And we assume that the matter is at rest in the comoving frame, which means that the 4-velocity is given by $u^\mu = (1, 0, 0, 0)$. Thus we have the only non-zero component of the EM tensor is:

$$T_{00} = \rho, \quad T_{ij} = p g_{ij} \quad \text{or} \quad T^{00} = \rho, \quad T^{ij} = p g^{ij} = p\frac{\gamma^{ij}}{a^2} \tag{11}$$

Generally, the homogeneity and isotropy of the universe requires that the EM tensor $\rho(t), p(t)$ only depends on time. This suits that the metric is also only depends on time.

Normally, we assume that there is a relation between the energy density and pressure, depend on the matter itself. Thus, the EM tensor has only one degree of freedom, which is the energy density ρ.

2.1.2 Friedmann Equations

Now we can plug in the curvature tensor and the EM tensor into the Einstein Field Equation, and we can get the 00 component of the Einstein Field Equation, which is called the Friedmann Equation:

Theorem 2.1 **Friedmann Equation**

The Friedmann Equation is given by the 00 component of the EFE of a rest perfect fluid in the comoving frame, which is given by:

$$\left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3} \tag{12}$$

Let's now count the variable and equations, usually we **fix a $\boldsymbol{k}$** for the model we want to consider. Then, the EM Tensor have 1 DoF $\rho(t)$, the metric have 1 DoF $a(t)$, and we have 1 equation, which is the Friedmann Equation. Thus we need another equation to solve the system.

We first consider the other Terms of the Einstein Field Equation, which is the ij component. We can get another equation from the ij component of the EFE, which is given by:

Theorem 2.2 **Spacial Component of EFE**

The ij component of the EFE of a rest perfect fluid in the comoving frame is given by:

$$2\frac{\ddot{a}}{a}+\left(\frac{\dot{a}}{a}\right)^2+\frac{k}{a^2}=-8\pi Gp+\Lambda \tag{13}$$

Remark

We only have one spacial equation, this is due to the fact that the spacial metric is dependent on γ_{ij} up to one factor. Thus, the spacial component of the EFE only gives one equation.

Sometimes we combine the two equations together, and we can get rid of the term $\dot{a}$, whic is :

$$\frac{\ddot{a}}{a}=-\frac{4\pi G}{3}(\rho+3p)+\frac{\Lambda}{3} \tag{14}$$

From this equation we can easily see when the universe is accelerating or decelerating. If we assume that $\Lambda=0$ then:

- If $\rho+3p>0$ then the universe is decelerating $\ddot{a}<0$
- If $\rho+3p<0$ then the universe is accelerating $\ddot{a}<0$

2.1.3 Energy Conservation of Matter

Another commonly used choice of the second equation is the energy conservation of matter. Note theta this equation is not independent of EFE, since its given by EFE and the Bianchi Identity.

Theorem 2.3 **Energy Conservation of Matter**

The energy conservation of matter is given by:

$$\partial_t(\rho a^3)+p\partial_t a^3=0 \tag{15}$$

We can see that the energy conservation is equivalent to the first law of thermodynamics, if we view the matter as a thermodynamic system with volume $V=a^3$ and internal energy $U=\rho a^3$.

2.2 Matter Content and Solutions

Now we try to find solutions to the EFE. But we need to specify the matter content exits in the universe.

2.2.1 Matter Content and Equation of State

We usually focus on Model behave as

$$P=\omega\rho \tag{16}$$

And usually the coefficient ω is given by:

- Pressureless Dust: $\omega=0$
- Relativistic Fluid (Radiation): $\omega=\frac{1}{3}$
- Cosmological Constant: $\omega=-1$

We believe that $\omega \geq -1$ and cc is the lower bound, which is called the null energy condition.

> **Remark**
>
> We can see from the EFE that a matter with $\omega = -1$ gives out $\rho = \text{const}$ in the energy consevation of matter, this is equivalent to having a cosmological constant of $\Lambda = 8\pi G\rho$.
>
> Btw, we may assume that $\rho > 0$ the matter have positive energy density. Thus, the "cosmological constant" case is in fact "positive cosmological constant" case.

2.2.2 Solution 1: Einstein Static Universe

If we assume that $\rho \neq 0,\ p = 0\ ;\ \dot{a} = 0$. which means that the universe is full of pressureless dust and is static. Then the two EFE becomes:

$$\frac{k}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3}, \quad \frac{k}{a^2} = \Lambda \tag{17}$$

If we take a positive cosmological constant, then we can get the following solution:

$$k = 1 \quad a = \sqrt{\frac{1}{\Lambda}} \quad \rho = \frac{\Lambda}{4\pi G} \tag{18}$$

This solution is called the Einstein Static Universe, which is a static universe with positive curvature and a positive cosmological constant. However, this is unrealistic

2.2.3 Solution 2: Flat Matter Dominated Universe

We assume that $k = 0$, $\Lambda = 0$, and the universe is full of pressureless dust. Then the Friedmann Equation and the Energy Conservation of Matter becomes:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho, \quad \partial_t(\rho a^3) = 0 \quad \text{or} \quad \rho a^3 = C \tag{19}$$

we can get a differential equation:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\frac{C}{a^3} \tag{20}$$

The solution to this equation is given by:

$$a(t) = a_0(t - t_0)^{\frac{2}{3}} \quad \rho \sim \frac{1}{t^2} \tag{21}$$

We can notice that this solution $\dot{a} > 0$ and $\ddot{a} < 0$, which means that the universe is expanding and decelerating.

2.2.4 Solution 3: Flat General Matter Dominated Universe

Consider a more general case with $k = 0$, $\Lambda = 0$, and the matter content is given by a general equation of state $P = \omega\rho$. This time the Friedmann Equation and the Energy Conservation of Matter becomes:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho, \quad \dot{\rho} + 3\left(\frac{\dot{a}}{a}\right)(\rho + p) = 0 \tag{22}$$

This gives us a solution:

$$\rho = Ca^{-3(1+\omega)}, \quad a(t) = a_0(t - t_0)^{\frac{2}{3(1+\omega)}} \tag{23}$$

There are some special properties of this solution:

- **Null Energy Condition** : Notice that we believe that $\omega \geq -1$. Thus we can see that $a(t)$ will always have a singularity at $t = t_0$, for the metric in this coordinate.
- **Matter density time dependent** :We can see that the if one kind of matter dominante the univers, then:

$$\rho \sim \frac{1}{t^2} \tag{24}$$

And if we have many kinds of matter and the effective ω is given by ω, then matter ρ_i evolve as:

$$\rho_i \sim t^{-2\frac{1+\omega_i}{1+\omega}} \tag{25}$$

- **Accelerating and Decelerating** : we can calculate $\ddot{a}$ and we see that:
  - If $\omega > -\frac{1}{3}$ then the universe is decelerating $\ddot{a} < 0$
  - If $\omega < -\frac{1}{3}$ then the universe is accelerating $\ddot{a} > 0$
- **Spacial Curvature as Matter**: An interesting observation is that having non-zero spacial curvature k is equivalent as choosing a matter of:

$$\omega = -\frac{1}{3} \tag{26}$$

  Thus in fact we can view the spacial curvature as a kind of matter content in the universe. We can also see that this matter makes the universe not accelerating nor decelerating, $\ddot{a} = \text{const}$.

3 Lecture 3: Cosmological Horizons, dS Spacetime and Distance

3.1 Cosmological Horizons

We consider the Penrose diagram of a FRW metricw, generally the metric can be written as:

$$ds^2 = -dt^2 + a(t)^2 dx^2 \tag{27}$$

Here we use dx^2 to represent the spactial metric. Then we can make a change of vairables:

$$d\eta = \frac{dt}{a(t)} \tag{28}$$

Thus we can have the metric using (η, x) coordinate as:

$$ds^2 = a(\eta)^2(-d\eta^2 + dx^2) \tag{29}$$

which is a conformally Minkowski metric. Then we need to fix the shape of the Penrose diagram. To do this we have to know what value can we take for η as t goes from $t_0 = 0$ to $t = +\infty$. We can integrate the relation Equation 28 and assume at $t = 0, \eta = \eta_*$:

$$\eta = \int_0^t \frac{dt'}{a(t')} + \eta_* \tag{30}$$

We assume that the space is flat and the spacetime is filled with $P = \omega\rho$ content. Then a general solution is:

$$a(t) \sim t^{\frac{2}{3(1+\omega)}} \tag{31}$$

Then we can see that $\omega > -\frac{1}{3}$ and $\omega < -\frac{1}{3}$ gives two cases for the behavior of η as t goes to zero and infinity.

3.1.1 Case 1: $\omega > -\frac{1}{3}$ decelerating universe

In the case, we can find that the integral Equation 30 converge at $t = 0$ but diverge at $t = +\infty$. we can choose a suitable η_* to make η go from η_* to $+\infty$, and at $t = 0, \eta = \eta_*$ the metric is singular. Thus the penrose diagram behave as:

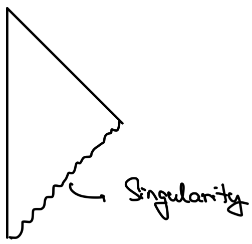

Figure 1: Penrose diagram for a decelerating universe with $\omega > -\frac{1}{3}$

YL: [This diagram may be wrong!!! check it??]

3.1.2 Case 2: $\omega < -\frac{1}{3}$ accelerating universe

In this case, we find the integral Equation 30 diverge at $t = 0$ but converge at $t = +\infty$. This means that for an accerlating universe, we can choose a suitable η_* to make η go from $-\infty$ to a finite number and at $t = 0, \eta = -\infty$ there is a singularity. Thus the penrose diagram be like:

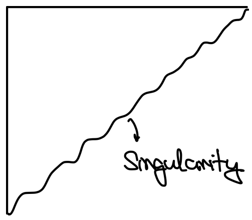

Figure 2: Penrose diagram for an accelerating universe with $\omega < -\frac{1}{3}$

We can see that in this case, there exist something called a **Cosmological Horizon**. It is originate from the fact that two points in the spacetime can never send information to each other, for the light can never reach each other before touching the future infinity.

This is interpret as the Universe is accerlating in expanding, thus the light can never catch up with the expansion of the universe.

3.2 De Sitter Spacetime

3.2.1 Cosmological Constant Case with Flat Spatial Curvature

We now focus on a special case of $k = 0, \Lambda = 0$ universe with the matter content of $\omega = -1$ which is the cosmological constant case. Or equivalently $\Lambda \neq 0, \rho = 0$. Note that it is not include in the general matter case Section 2.2.4, since the solution is distinct.

In this case, the Friedmann Equation and Energy Conservation of Matter becomes:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{\Lambda}{3}, \quad \rho = \text{const} := \frac{\Lambda}{8\pi G} \tag{32}$$

Thus we can get the solution that:

$$a(t) = \exp(Ht), \quad H = \sqrt{\frac{\Lambda}{3}}, \tag{33}$$

This solution we can see that t goes from $-\infty$ to $+\infty$. There is a singularity at $t = -\infty$ where $a(t) = 0$.

> **Remark**
>
> Here we take a convention that $a_0 = 1$ and in fact there are two solutions e^{Ht} and e^{-Ht}. We can choose the solution e^{Ht} since it is more physical, which means that the universe is expanding. The other solution e^{-Ht} is a contracting universe, and singularity at $t = +\infty$ where $a(t) = 0$. However, this solution is not physical since we believe that the universe is expanding, thus we can discard this solution.

3.2.2 Penrose Diagram

We can make a change of variable:

$$\eta = \int \frac{dt}{a(t)} = -\frac{e^{-Ht}}{H} \tag{34}$$

Thus we can get the metric in the (η, x) then the metric becomes:

$$ds^2 = \frac{a_0^2}{H^2\eta^2}(-d\eta^2 + dx^2) \tag{35}$$

we can see the range of the parameter η:

- as $t \to \infty$ $\eta \to 0^-$ which is far future.
- as $t \to -\infty$ $\eta \to -\infty$ which is the past singularity.

Thus the penrose diagram of this spacetime is like:

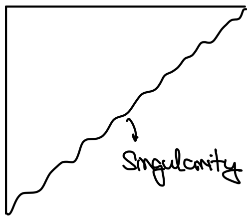

Figure 3: Penrose diagram for cosmological constant case with $\omega < -1$

which behaves just like the accerlating universe case in the general matter solution Figure 2.

3.2.3 Global de Sitter Spacetime

However, the singularity we get at $t = -\infty$ is not a real singularity!! These can be seen by the behavior of the curvature. We can prove that the curvature is a constant and don't have any singularity at $t = -\infty$. Thus, we expect we can do an extension of the spacetime to get rid of this coordinate singularity.

One approach to this is to investigate a seemingly different solution of the EFE, which is the **vaccum solution with $k = 1$**. The EFE reads:

$$\left(\frac{\dot{a}}{a}\right)^2 + \frac{1}{a^2} = \frac{\Lambda}{3}, \quad 2\left(\frac{\ddot{a}}{a}\right) + \left(\frac{\dot{a}}{a}\right)^2 + \frac{1}{a^2} = \Lambda \tag{36}$$

The solution to this equation is given by:

$$a(t) = H^{-1}\cosh(Ht), \quad H = \sqrt{\frac{\Lambda}{3}}, \tag{37}$$

This spacetime is called **Global dS Spacetime**.

$$ds^2 = -dt^2 + H^{-2}\cosh^2(Ht)(d\chi^2 + \sin^2(\chi)d\Omega_2^2) \tag{38}$$

We can see that the metric is non-singular at any t for the cosh function is well behaved and have no zero point. More importantly, we can prove:

- The cosmological constant case with flat spatial curvature is a patch of the global dS spacetime, which is called the **Poincare Patch**.

In penrose diagram, this can be shown as:

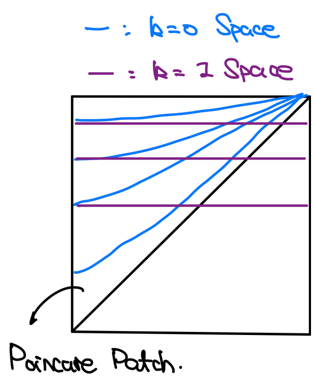

Figure 4: Penrose diagram for the global dS spacetime, where the blue region is the Poincare patch which is the cosmological constant case with flat spatial curvature

We can see that the blue line is a spacial slicing that make the spacial metric flat, yet the price to pay is that it only cover a patch of the global dS spacetime.

3.3 Hubble Parameter and General Evolution of the Universe

We recap on the three equations we commonly use for solving EFE:

$$\begin{aligned} &\left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3}, \quad \text{Friedmann Euqation} \\ &2\left(\frac{\ddot{a}}{a}\right) + \left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} = -8\pi G p + \Lambda, \quad \text{Spacial EFE} \\ &\dot{\rho} + 3\left(\frac{\dot{a}}{a}\right)(\rho + p) = 0 \quad \text{Energy Conservation of Matter} \end{aligned} \tag{39}$$

If we consider a certain kind of matter $p_i = \omega_i \rho_i$, the enegy conservation gives us:

$$\rho_i = C a^{-3(1+\omega_i)}, \quad p_i = \omega_i \rho_i \tag{40}$$

Moreover, we can define a parameter:

Definition 3.1 **Hubble Parameter**

The **Hubble Parameter** is defined as:

$$H = \frac{\dot{a}}{a} \tag{41}$$

Thus the Friedmann Equation with $k = 0, \Lambda = 0$ can be written as:

$$H^2 = \frac{8\pi G}{3} \sum_i \rho_i \tag{42}$$

Further more we can define a parameter called **Critical Density** if we use the energy density of **today** as a reference:

Definition 3.2 **Critical Density**

The **Critical Density** is defined as:

$$\rho_c = \frac{3H_0^2}{8\pi G}, \quad \text{equivalently} \quad \rho_c = \sum_i \rho_i(t = t_0) \tag{43}$$

where H_0 is the Hubble Parameter at present time t_0. And we can define Ω_i which is :

$$\Omega_i = \frac{\rho_i(t = t_0)}{\rho_c}, \quad \text{thus} \quad \sum_i \Omega_i = 1 \tag{44}$$

With all these preparation, we can rewrite the Friedmann Equation (with $k \neq 0, \Lambda \neq 0$ and with the energy conservation already used) as:

$$H^2 = H_0^2 \left(\Omega_\Lambda + \Omega_k \left(\frac{a_0}{a}\right)^2 + \Omega_m \left(\frac{a_0}{a}\right)^3 + \Omega_\gamma \left(\frac{a_0}{a}\right)^4 \right) \tag{45}$$

The first two term is the contribution from the cosmological constant and the spatial curvature, and the last two term is the contribution from the matter and the radiation, with certain ω_i determine its behavior.

If we rewrite LHS back with $\frac{\dot{a}}{a}$ and we group terms we can get:

$$\dot{a}^2 = H_0^2 \left(\Omega_\Lambda a^2 + \Omega_k a_0^2 + \Omega_m \frac{a_0^3}{a} + \Omega_\gamma \frac{a_0^4}{a^2} \right) \tag{46}$$

Which can be interpreted as the equation of motion of a particle with zero total energy moving in a potential:

$$U(a) = -H_0^2 \left(\Omega_\Lambda a^2 + \Omega_k a_0^2 + \Omega_m \frac{a_0^3}{a} + \Omega_\gamma \frac{a_0^4}{a^2} \right), \quad \dot{a}^2 + U(a) = 0 \tag{47}$$

Why do we prefer this form of the EoM of the metric?

This is because prameters here are mostly observables that we can measure from looking into the ski.

- H_0 can be measured through measuring how stars go away from us. And therefore we can measure the critical density ρ_c.
- Ω_i can be measured by doing sky search, and seeing the energy density of different matter content in the universe.

3.4 Distances in Cosmology

To investigate the cosmology, we need to define the lengths.

3.4.1 Redshift

Assume a process, a star emit a light at $t = t_1, r = r_1$ and the light is observed by us at $t = t_0, r = 0$. The light travel along null geodesics:

$$ds^2 = -dt^2 + a(t)^2 \frac{dr^2}{1 - kr^2} = 0 \quad \Rightarrow \quad dt = -a(t) \frac{dr}{\sqrt{1 - kr^2}} \tag{48}$$

There is a "-" sign since the light is traveling from r_1 to 0. Thus we can get evaluate a length of the light travel l(r):

$$\int_{t_1}^{t_0} \frac{dt}{a(t)} = \int_0^{r_1} \frac{dr}{\sqrt{1 - kr^2}} = l(r_1) \tag{49}$$

We can calculate the length l(r) for different spatial curvature:

$$l(r) = \begin{cases} \operatorname{arcsin}(r) & k = 1 \\ r & k = 0 \\ \operatorname{arcsinh}(r) & k = -1 \end{cases} \tag{50}$$

Then we can consider a process of two light signal emit at $t = t_1$ and $t = t_1 + \delta t_1$ and the light is observed at $t = t_0$ and $t = t_0 + \delta t_0$. Since the two light signal travel along the same null geodesics, we can get:

$$\frac{\delta t_0}{a(t_0)} = \frac{\delta t_1}{a(t_1)} \tag{51}$$

The frequency of the light is inversely proportional to the time interval $\omega \propto \frac{1}{t}$, thus we can get:

$$\frac{\omega_0}{\omega_1} = \frac{\delta t_1}{\delta t_0} = \frac{a(t_1)}{a(t_0)} \tag{52}$$

- Universe is Expanding: $a(t_0) > a(t_1)$ thus $\omega_0 < \omega_1$ which is called **Redshift**.

- Universe is Contracting: $a(t_0) < a(t_1)$ thus $\omega_0 > \omega_1$ which is called **Blueshift**.

The redshift can be described by the following quantity:

Definition 3.3 **Redshift**

The **Redshift** is defined as:

$$z = \frac{\lambda_0 - \lambda_1}{\lambda_1} = \frac{a(t_0)}{a(t_1)} - 1 \tag{53}$$

note that t_0 is the time of getting the light and t_1 is the time of emitting the light.

Usually, we know the λ_1 of many processes, and we measure λ_0 from experiments. Thus, we can use z as a measure of the distance.

3.4.2 Physical Distance and Coordinate Distance

For two comoving objects at $r = r_1$ and $r = r_2$, the coordinate distance is:

$$l_c = l(r_1) - l(r_2) \tag{54}$$

which is the distance separated in the spacial coordinate. However, its not the distance in the general metric. The physical distance is given by:

$$l_p = a(t)(l(r_1) - l(r_2)) = a(t)l_c \tag{55}$$

We know that for a comoving object, the spacial coordinate is fixed when moving in geodesics. Yet, the physical distance is time dependent due to the $a(t)$ parameter:

$$\frac{dl_p}{dt} = \dot{a}l_c = Hl_p, \quad H = \frac{\dot{a}}{a} \tag{56}$$

This is the **Hubble Law**, which fits well with the observation of the universe, which is one of the evidences for the expanding universe.

Remark

We here have three notion of "length":

- r, the spacial coordinate variable, which is NOT the coordinate distance.
- $l(r)$, the coordinate distance, which is the distance in the spacial metric.
- l_p, the physical distance, which is the distance in the full metric.

Note

It is also important to note that we assume the bodies in the universe have fixed co-moving coordinates in space, which means that the coordinate distance and spacial coordinates are all constants. However, due to the expansion of the universe the physical distance is time dependent.

3.4.3 Luminosity Distance

We first define something as:

- **Standard Candle** : which is a object with known luminosity L, where $L = \frac{dE}{dt}$ is defined as the energy emitted by the object per unit time in the object's rest frame.

Then we consider a comoving telescope observing the standard candle, we can define a quantity called **Luminosity Distance** as:

Definition 3.4 **Luminosity Distance**

The **Luminosity Distance** is defined as:

$$P = L\frac{S}{4\pi d^2} \tag{57}$$

where P is the measured energy recieved per unit time by the telescope, and S is the area of the telescope. The luminosity distance d is defined as the distance that make the above equation hold.

Then we can calculate the luminosity distance and its relation with other quantities for a comoving telescope and star in the FRW metric set up.

In the FRW metric set up, the actual P can be calculated as:

$$P = L\left(\frac{S}{S_{\text{tot}}}\right)\left(\frac{\hbar\omega_0}{\hbar\omega_1}\right)\left(\frac{\delta t_1}{\delta t_0}\right) \tag{58}$$

Lets explain each term, we set convension of t_1 the time of emitting the light and t_0 the time of getting the light:

- L is the Energy emitted per unit time when the energy is emitted.
- S is the area of the telescope. S_{tot} is the total area of light front, due to the expansion of the universe it takes form of:

$$S_{\text{tot}} = 4\pi a(t_0)^2 r^2(t_1, t_0) \tag{59}$$

where $r(t_1, t_0)$ is the difference in coordinate r between the star and the telescope (which is NOT the coordinate distance), which follow the null geodesic equation as:

$$dt = -a(t)\frac{dr}{\sqrt{1-kr^2}} \Rightarrow \int_{t_1}^{t_0}\frac{dt}{a(t)} = \int_0^r \frac{dr}{\sqrt{1-kr^2}} \tag{60}$$

where r then is implicitly a function of t_1 and t_0.

Remark

Why do this S_{tot} is the total area of the light front?

We can see this from the FRW metric Definition 1.2, for a fixed time t_0, and a fixed r, the area of the sphere is given by:

$$ds^2 = a^2(t_0)r^2 d\Omega_2^2 \tag{61}$$

we integrate over the angular coodiantes we can get the area of the sphere as:

$$S = 4\pi a^2(t_0)r^2 \tag{62}$$

- $\frac{\omega_0}{\omega_1}$ is the ratio of the frequency of the light when getting and emitting, which is given by the redshift as:

$$\frac{\omega_0}{\omega_1} = \frac{a(t_1)}{a(t_0)} \tag{63}$$

- $\frac{\delta t_1}{\delta t_0}$ is the time given by the difference of infinitesimal time changes. This is also given by the redshift, see Equation 51:

$$\frac{\delta t_1}{\delta t_0} = \frac{a(t_1)}{a(t_0)} \tag{64}$$

If we put all these together, we can get:

$$P = L\left(\frac{S}{S_{\text{tot}}}\right)\left(\frac{\hbar\omega_0}{\hbar\omega_1}\right)\left(\frac{\delta t_1}{\delta t_0}\right) = L\left(\frac{S}{4\pi a(t_0)^2 r^2}\right)\left(\frac{a(t_1)}{a(t_0)}\right)^2 = LS\frac{a(t_1)^2}{4\pi a(t_0)^4 r^2} \tag{65}$$

This gives us the luminosity distance as:

Corollary 3.5 **Luminoity Distance in FRW Metric**

The luminosity distance in the FRW metric is given by:

$$d = \frac{a(t_0)^2}{a(t_1)} r(t_1, t_0) = (1+z)a(t_0)r(t_1, t_0) \tag{66}$$

where z is the redshift, see Definition 3.3.

3.5 Distances in Cosmology II

3.5.1 Luminosity Distance and Redshift

We want a formula for the luminosity distance d_L as a function of redshift z. Thus, if we both measure d_L and z then we can find measure parameter $a(t)$.

We start from the expression Corollary 3.5, we can see that $r(t_0, t_1)$ can be calculated using the null geodesic Equation 60, using the result of Equation 50 gives:

$$r(t_0, t_1) = \begin{cases} \int_{t_1}^{t_0}\left(\frac{dt}{a(t)}\right) & k = 0 \\ \sin\left(\int_{t_1}^{t_0}\left(\frac{dt}{a(t)}\right)\right) & k = 1 \\ \sinh\left(\int_{t_1}^{t_0}\left(\frac{dt}{a(t)}\right)\right) & k = -1 \end{cases} \tag{67}$$

Then we want to calculate the integral:

$$\int_{t_1}^{t_0}\left(\frac{dt}{a(t)}\right) \tag{68}$$

For a general universe, the Friedmann equation and Energy conservation equation gives Equation 45. We copy it here:

$$H^2 = H_0^2\left(\Omega_\Lambda + \Omega_k\left(\frac{a_0}{a}\right)^2 + \Omega_m\left(\frac{a_0}{a}\right)^3 + \Omega_\gamma\left(\frac{a_0}{a}\right)^4\right) \tag{69}$$

Remark

Note that in this equation we set a time t_0 to define H_0 ,a_0 and Ω_i these constants. Here in the context of distance, we take t_0 to be the time when the light is emitted (note this conficted with the previous convention where we take t_0 to be the time when the light is observed, and t_1 to be the time when the light is emitted).

We make a change of variable $x = \frac{a}{a_0}$, then we have:

$$\dot{x} = H_0 A(x) x, \quad A(x)^2 = \left(\Omega_\Lambda + \Omega_k x^{-2} + \Omega_m x^{-3} + \Omega_\gamma x^{-4}\right) \tag{70}$$

With these preparation we can calculate the integral:

$$\int_{t_1}^{t_0} \left(\frac{dt}{a(t)} \right) = \frac{1}{a_0 H_0} \int_{x_1}^{x_0} \left(\frac{dx}{A(x)x^2} \right) \quad x_0 = 1, x_1 = \frac{a(t_1)}{a_0} = \frac{1}{1+z} \tag{71}$$

Then the final formula for the luminosity distance is given by:

Corollary 3.6 **Luminosity Distance ito Redshift**

The luminosity distance can be calculated as a function of redshift as:

$$d(z) = a_0(1+z)\frac{1}{\sqrt{-k}} \sinh\left(\frac{\sqrt{-k}}{a_0 H_0} \int_{\frac{1}{1+z}}^{1} \left(\frac{dx}{A(x)x^2} \right) \right) \tag{72}$$

This is combining the three cases of k into one formula, for $k = 0$ we take the limit $k \to 0$ and use the fact that $\lim_{x\to 0}\left(\frac{\sinh(x)}{x}\right) = 1$.

3.5.2 Parallax

Consider we have a source very far from the sun and we want to measure the distance between another star, which is much closer to the sun (though still very far from us). We can measure the angle between the source and the star at two different time (for example, at two different position of the earth in its orbit around the sun). This is called parallax.

See the diagram for illustration:

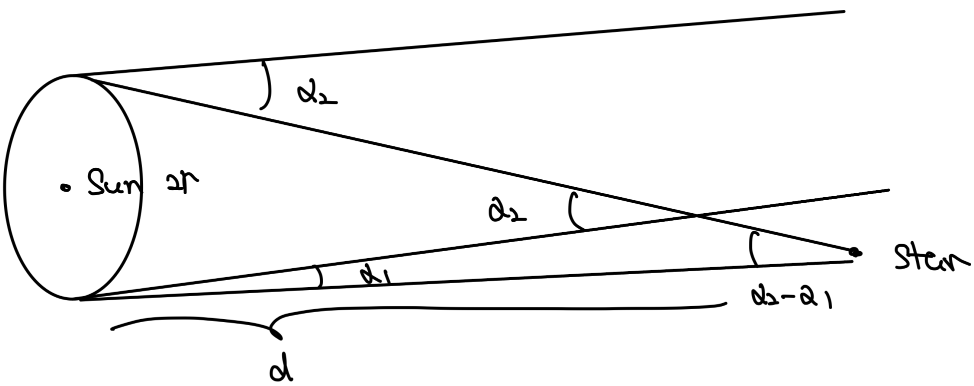

Figure 5: Parallax measurement.

The parallax is defined as:

Definition 3.7 **Parallax**

The parallax is defined as the angle between the source and the star at two different time.

$$\alpha_2 - \alpha_1 \tag{73}$$

A distance can be calculated from the parallax as:

$$d = \frac{2r}{\alpha_2 - \alpha_1} \tag{74}$$

where r is the distance between the sun and the earth.

4 Lecture 4: Thermal History of the Universe I: Equilibrium

4.1 Review on Euquilibrium Statistical Mechanics

4.1.1 Occupation Number and Thermal Distribution Function

For a Equilibrium system, Bosons and Fermions have different occupational number:

Theorem 4.1 **Occupartion Number of Bosons and Fermions**

For a system in equilibrium, the occupation number of bosons and fermions are given by:

$$n_B(E) = \frac{1}{e^{\frac{E-\mu}{T}} - 1}, \quad n_F(E) = \frac{1}{e^{\frac{E-\mu}{T}} + 1} \tag{75}$$

where E is the energy of the state, μ is the chemical potential and T is the temperature.

The thermal distribution function of the particles are given by $f(p)$, it is related to the occupation number and the number of dof of the particle:

Definition 4.2 **Thermal Distribution Function**

The thermal distribution function of the particle i is given by:

$$f_i(p) = \left(\frac{g_i}{(2\pi)^3}\right) n_i(E(p)) \quad E(p) = \sqrt{p^2 + m_i^2} \tag{76}$$

Remark

As a reminder of convention of statistical mechanics. the $(2\pi\hbar)^3$ or in the above formula $(2\pi)^3$ where we set $\hbar = 1$ is the density of phase space. and g_i is the number of degrees of freedom of the particle, for example, for a photon we have $g_i = 2$ because it has two polarization states.

For total number of partitcle, we need to integrate over the phase space d^3xd^3p, yet we leave the spacial part d^3x out because we are interested in the number density, which is the number of particle per unit volume. Thus we only integrate over the momentum space d^3p.

Integrating over $f_i(p)$ gives the **number density** of the particle and integrating with the weight of energy gives the **energy density** of the particle:

- **Number Density**:

$$N_i = \int f_i(p) d^3p \tag{77}$$

- **Energy Density**:

$$\rho_i = \int E(p) f_i(p) d^3p \tag{78}$$

4.1.2 Energy and Number Density in Relativistic Limit

The number density and energy density at equilibrium can be calculated explicitly. For particles in the **relativistic limit**, we means really high temperature::

$$T \gg m_i \quad \Rightarrow \quad T \sim E \sim |p| \tag{79}$$

and now we assume **zero chemical potential** we can caluclate the number density and energy density as:

- **Energy Density**:

$$\rho_i = \begin{cases} \left(\frac{\pi^2}{30}\right) g_i T^4 & \text{Boson} \\ \\ \frac{7}{8}\left(\frac{\pi^2}{30}\right) g_i T^4 & \text{Fermion} \end{cases} \tag{80}$$

- **Number Density**:

$$N_i = \begin{cases} \left(\frac{\zeta(3)}{\pi^2}\right) g_i T^3 & \text{Boson} \\ \\ \frac{3}{4}\left(\frac{\zeta(3)}{\pi^2}\right) g_i T^3 & \text{Fermion} \end{cases} \tag{81}$$

We can also calculate the average energy of the particle in this limit:

$$\langle E \rangle = \frac{\rho_i}{N_i} \tag{82}$$

For the convenience of calculating energy density of system with multiple particle species, we can define the **effective number of species**:

$$g_* = \sum_{\text{boson}} g_i + \left(\frac{7}{8}\right) \sum_{\text{fermion}} g_i \tag{83}$$

and the energy density of the system can be calculated as:

$$\rho = \left(\frac{\pi^2}{30}\right) g_* T^4 \tag{84}$$

4.1.3 Entropy Density in Relativistic Limit

From the second law of thermodynamics, we have:

$$E = TS - PV + \mu N \tag{85}$$

we take the assumption that the chemical potential is zero, then we have:

$$s = \frac{S}{V} = \frac{\rho + P}{T} \tag{86}$$

where ρ is the energy density and P is the pressure of the system. For a relativistic particle remeber we have Equation 16, we take $P = \frac{1}{3}\rho$ for relativistic particle, thus we have:

$$s = \frac{4}{3}\left(\frac{\rho}{T}\right) \tag{87}$$

4.1.4 Energy and Number Density in Non-Relativistic Limit

For non-relativistic limit, we means that:

$$m_i \gg T \quad \Rightarrow \quad E \sim m_i + \frac{p^2}{2m_i} \tag{88}$$

Note that here μ can be important. In this region the number density and energy density can be calculated as:

- **Number Density**:

$$N_i = g_i \left(\frac{m_i T}{2\pi} \right)^{\frac{3}{2}} \exp\left(\frac{\mu_i - m_i}{T} \right) \tag{89}$$

- **Energy Density**:

$$\rho_i = m_i N_i(T) + \frac{3}{2} N_i T \tag{90}$$

However, due to the fact that its a non-relativistic particle, the rest mass energy is much larger than the kinetic energy, thus we can approximate the energy density as:

$$\rho_i = m_i N_i \tag{91}$$

Moreover the pressure of the particle is given by:

$$P_i = g_i N_i T \ll \rho_i \tag{92}$$

Thus, we can see Equation 16 that the pressureless dust approximation is valid for non-relativistic particle.

4.2 Temperature Today

With these preparation of thermaldynamics, we can calculate the temperature today. We first assume that the universe is:

- $k = 0, \Lambda = 0$ and dominated by relativistic particles that are in thermal equilibrium.

then the Friedmann equation gives Equation 42 :

$$H = \left(\frac{8\pi G}{3} \rho_{\text{rad}} \right)^{\frac{1}{2}} \tag{93}$$

And from the thermaldynamics calculation we have Equation 84, we then can see the temperature dependency of the Hubble parameter:

$$H = \frac{T^2}{M_0} \quad M_0 = \sqrt{\frac{45}{4\pi^3 G g_*}} \tag{94}$$

where g_* is the effective number of species at the time.

For a radiation dominated universe, we have $a(t) \sim t^{\frac{1}{2}}$, or if we choose a time $t_0$to define a_0 we have $a(t) = a_0 \left(\frac{t}{t_0} \right)^{\frac{1}{2}}$. Then:

$$H = \frac{\dot{a}}{a} = \frac{1}{2t} \tag{95}$$

Finally, we arrive at the relation between temperature and time:

$$\frac{T^2}{M_0} = \frac{1}{2t} \quad \Rightarrow \quad T = \sqrt{\frac{M_0}{2t}} \tag{96}$$

Now we can esitimate with real numbers. for g_* we can plug in the standard model particles which gives $g_* \sim 100$. If we plug in the age of the universe, then the temperature turns out to be quite as the same of CMB.

The calculated temperature is not really the CMB, but a temperature that we assume the universe has if its always radiation dominant and in thermal equilibrium. However, in fact:

1. Today the universe is not radiation dominated.

2. The CMB temperature is the freeze-out temperature of the photon, which is the effective temperature after the photon decouples from the matter and stop interacting with it.

Yet its quite amazing that this really rough calculation can give us a quite sensible result.

Here we can also draw a useful relation between $a(t)$ and T, for a radiation dominated universe and roughly assuming the thermal equilibrium, we have:

$$\frac{a(t)}{a_0} = \left(\frac{t}{t_0}\right)^{\frac{1}{2}} = \left(\frac{T_0}{T}\right) \tag{97}$$

5 Lecture 5: Thermal History of the Universe II: Non-Equilibrium

The Universe is not always in thermal equilibrium, now we have to consider the non-equilibrium processes in the universe.

5.1 Free Particle Boltzmann Equation

5.1.1 A Convention Shift

For a non-equilibrium system, we need to consider the Boltzmann equation, which describes the evolution of the distribution function of particles in phase space. Here we use a different convention (as in the lecture note), we have:

$$n_{i\ \text{old}}(p,t) = \frac{(2\pi)^3}{g_i} f_i(p,t) \quad \rightarrow \quad n_i(p,t) = (2\pi)^3 f_i(p,t) \tag{98}$$

Then the number density is:

$$N_i = \int \frac{d^3p}{(2\pi)^3} n_i(p,t) \tag{99}$$

and similarly for the energy density.

An important remark is that the distribution p we use the **physial momentum**, for detailed definition see next subsection.

5.1.2 Free Particle in an Expanding Universe

We first consider the free particle, in an expanding universe. The Lagrangian is given by:

$$\mathcal{L} = \frac{1}{2} m g_{ij} (\dot{x}^i \dot{x}^j) \tag{100}$$

here $\dot{x}$ is the derivative with respect to the coordinate time t. We now can explicitly define two momentums:

- **Physical Momentum**: $\mathcal{L} = \frac{p_p^2}{2m}$ this gives us: $p_p^i = m a(t) \dot{x}^i$.
- **Comoving Momentum**: $p_c^i = \frac{\partial \mathcal{L}}{\partial \dot{x}^i}$ this gives us: $p_c^i = m a(t)^2 \dot{x}^i$.

Then from the Euler-Lagrange equation, we can get the equation of motion of the particle, which is given by:

$$\frac{d}{dt} p_c^i = 0 \tag{101}$$

> **Remark**
>
> This can also be view given by the Noether's theorem of spacial translation symmetry.

Wethen can see that the comoving momentum is conserved, which means that the physical momentum is redshifted by the expansion of the universe:

$$p_p^i = \frac{p_c^i}{a(t)} = \frac{\text{constant}}{a(t)} \tag{102}$$

if we choose an "initial time" t_0 and define the physical momentum at that time as $p_0$and $a(t_0) = a_0$, then we can write the physical momentum at any time as:

$$p_0^i = \frac{p_c^i}{a_0} = p_p^i \frac{a(t)}{a_0} \tag{103}$$

Thus given a time t, we want to measure the number of particle with physical momentum p, due to the fact that the comoving momentum is conserved, the number is equal to number of particle at t_0 with physical momentum $p\frac{a(t)}{a_0}$. Thus we have:

$$n_i(p,t) = n_i\left(p_0 = p\frac{a(t)}{a_0}, t_0\right) \equiv n_{0i}\left(p\frac{a(t)}{a_0}\right) \tag{104}$$

Why we care about this form of distribution depends on some "initial distribution"? it is because particles in the universe, they first interact with each other when temperature is high and they are likely to stay in a thermal equilibrium distribution. However, when temperature lowers down, the interaction is not likely to happen, thus the particle will behave like a free particle and evolve.

This procesure ensures us the knowledge of the distribution at the initial time t_0 (for example a thermal equilibrium distribution) and we can use the above formula to see how the distribution evolves with time.

See the diagram for illustration:

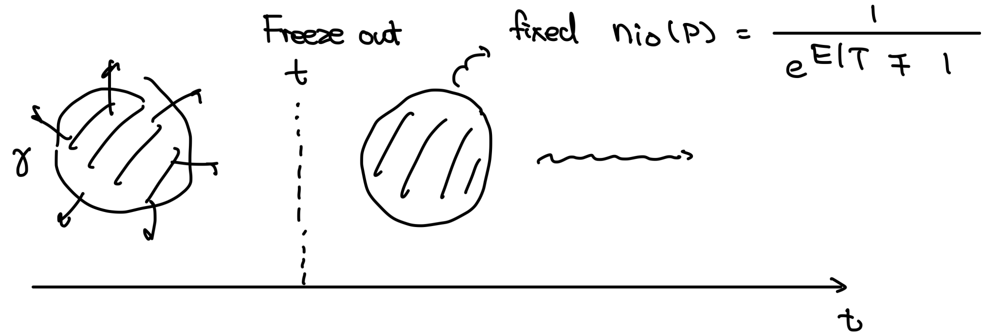

Figure 6: The process of freeze-out.

5.1.3 Effective Temperature

Now consider the process described in above remark. Initially the particle is in thermal equilibrium, thus we have a thermal distribution at the initial time t_0:

$$n_{i0}(p) = g_i \frac{1}{\exp\left(\frac{E(p)-\mu_i}{T_0}\right) \pm 1} \tag{105}$$

Then the interaction turns off and the particle evolves as a free particle, thus we have:

$$n_i(p,t) = n_{i0}\left(p\frac{a(t)}{a_0}\right) = g_i \frac{1}{\exp\left(\frac{E\left(p\frac{a(t)}{a_0}\right)-\mu_i}{T_0}\right) \pm 1} \tag{106}$$

In some cases, we can define an **effective temperature** and make all the time dependence of this distribution into the effective temperature. Thus we can use this effective temperature and a equilibrium distribution to describe the distribution of the particle at any time.

Remark

The effective temperature is not really a temperature for the particle is free and not in thermal equilibrium and we don't have a notion of temperature.

Yet it is a measurable quantity, for we can measure the energy distribution of the particle and fit it with a thermal distribution, then we can get the effective temperature from experiment.

- **Massless Particle**

For massless particle, the energy is given by $E(p) = p$, if we negelect the chemical potential, we have:

$$n_i(p,t) = g_i \frac{1}{\exp\left(\frac{pa(t)}{a_0 T_0}\right) \pm 1} \tag{107}$$

Then we define the effective temperature as:

Definition 5.1 **Effective Temperature for Massless Particle**

The effective temperature for a massless particle is given by:

$$T_{\text{eff}}(t) = T_0 \frac{a_0}{a(t)} \tag{108}$$

Then the distribution of the particle can be written as:

$$n_i(p,t) = g_i \frac{1}{\exp\left(\frac{p}{T_{\text{eff}(t)}}\right) \pm 1} \tag{109}$$

In fact in cosmology, we often use the mixed use the effective temperature of relativistic particle and the assumed equilibrium temperature of radiation dominant universe. A justification may be that they both evolve as:

$$T_{\text{eff}}(t) = T_0 \frac{a_0}{a(t)} \quad \text{and} \quad T(t) = T_0 \frac{a_0}{a(t)} \tag{110}$$

It is invalid rigorously I know, yet it amazingly works for many cases. For example, before we calculate the temperature of the universe as a theraml equilibrium relativistic system and the temperature turns out to be quite as the same of CMB which is the effective temperature of the photons in the universe.

However, sometimes it can't give a correct result and we need some further justifications. Eg. the difference between the effective temperature of neutrinos and the CMB temperature. This happens due to some extra processes happening in the universe that makes the evolution of temperature not that simple.

- **Massive Particle**

See section 2.5 of the book [1] for more details.

5.1.4 Free Particle Boltzmann Equation

Then we can derive the Boltzmann equation for a free particle in an expanding universe. We have:

$$\frac{\partial}{\partial t} n_i(p,t) = \frac{d}{dt} n_{0i} \left(p \frac{a(t)}{a_0} \right) = (n'_{0i}) \times \left(p \frac{\dot{a}(t)}{a_0} \right) \tag{111}$$

and

$$\frac{\partial}{\partial p} n_i(p,t) = n'_{0i} \times \left(\frac{a(t)}{a_0} \right) \tag{112}$$

we combine the two equations together, we can get:

$$\frac{\partial}{\partial t} n_i(p,t) - \left(\frac{\dot{a}(t)}{a(t)} p \right) \frac{\partial}{\partial p} n_i(p,t) = 0 \tag{113}$$

Then we have the Boltzmann equation for a free particle in an expanding universe:

Theorem 5.2 **Free Particle Boltzmann Equation**

The Boltzmann equation for a free particle in an expanding universe is given by:

$$\frac{\partial}{\partial t} n_i(p,t) - (Hp) \frac{\partial}{\partial p} n_i(p,t) = 0 \tag{114}$$

where $H = \frac{\dot{a}(t)}{a(t)}$ is the Hubble parameter.

We can see that a "free particle" in an expanding universe is not really free, it is affected by the expansion of the universe, which causes the redshift of the physical momentum.

We then can integrate the Boltzmann equation over the momentum space, we can get the equation for the number density:

$$\dot{N}_i(t) + 3H(t)N_i(t) = 0 \tag{115}$$

Remark

Remember the convention is changed and we integrate with $(d^3p)/(2\pi)^3$ measure. We'd mainly prefer using number density instead of distribution function as the variable.

5.2 Interacting Particle Boltzmann Equation

5.2.1 Interacting Particle Boltzmann Equation

Then we consider a particles with interactions. The generalization is done by adding a collision term I_{col} to the Boltzmann equation:

Theorem 5.3 **Interacting Particle Boltzmann Equation**

The Boltzmann equation for an interacting particle in an expanding universe is given by:

$$\frac{\partial}{\partial t} n_i(p,t) - (Hp) \frac{\partial}{\partial p} n_i(p,t) = I_{\text{col}} \tag{116}$$

where I_{col} is the collision term, which describes the interactions of the particle with other particles.

If we focus on **2 to 2 scattering procedure**, the collision term can be written in terms ofthe scattering amplitude of the procedure, which is given by:

$$
\begin{aligned}
I_{\text{col}} = -\frac{1}{2E_p} \int \frac{d^3q_1}{(2\pi)^3} \frac{d^3q_2}{(2\pi)^3} \frac{d^3q_3}{(2\pi)^3} (2\pi)^4 \delta^{(4)}(p + q_1 - q_2 - q_3)\ |\mathcal{M}|^2 \\
\times [n(p)n(q_1)(1 \pm n(q_2))(1 \pm n(q_3)) - n(q_3)n(q_2)(1 \pm n(p))(1 \pm n(q_1))]
\end{aligned}
\tag{117}
$$

This equation is quite imposible to solve analytically, yet we have the technique of **Relaxation Time Approximation**, which we expand around the equilibrium distribution $n_i^{\text{eq}}(p)$.

5.2.2 Relaxation Time Approximation

We consider at $H = 0$ we have the equilibrium distribution $n_i^{\text{eq}}(p)$, we can prove that:

$$
I_{\text{col}}(n_i^{\text{eq}}(p)) = 0 \quad \text{and} \quad \frac{d}{dt} n_i^{\text{eq}}(p) = 0 \tag{118}
$$

where n_i^{eq} is exactly the Bose-Einstein distribution or the Fermi-Dirac distribution.

- This means that in an interaction dominat system, all particle tends to have the equilibrium distribution, which is given by the Bose-Einstein distribution or the Fermi-Dirac distribution.

Then we can expand the actural distribution around the equilibrium distribution, in this limit:

$$
I_{\text{col}}(n_i) = I_{\text{col}}(n_i^{\text{eq}} + \delta n_i) = I_{\text{col}}(n_i^{\text{eq}}) + \left(\frac{\delta I_{\text{col}}}{\delta n_i} \right) \delta n_i + \ldots = -\Gamma \times (n_i - n_i^{\text{eq}}) \tag{119}
$$

Thus the Boltzmann equation becomes:

$$
\frac{\partial}{\partial t} n_i(p, t) - (Hp) \frac{\partial}{\partial p} n_i(p, t) = -\Gamma \times (n_i - n_i^{\text{eq}}) \tag{120}
$$

or if we integrate over the momentum space, we can get the equation for the number density:

Theorem 5.4 **Relaxation Time Approximation Boltzmann Equation**

The Boltzmann equation for an interacting particle in an expanding universe in the relaxation time approximation is given by:

$$
\dot{N}_i(t) + 3H(t)N_i(t) = -\Gamma \times (N_i(t) - N_i^{\text{eq}}) \tag{121}
$$

Now we talk more about Γ. It has a physical meaning as the **rate of reaction**:

$$
\Gamma = \frac{1}{\tau} = \langle \lambda / v \rangle^{-1} \tag{122}
$$

where τ is the **mean free time** of the collition and λ is the **mean free path** of the collition and v is the **relative velocity** of the particle. The mean free path is related to the **cross section** of the collition by:

$$
\lambda = \frac{1}{\sigma N} \tag{123}
$$

where σ is the cross section of the collition and N is the number density of the particle. A commonly used formula for Γ is given by:

$$
\Gamma = \langle N v \sigma \rangle \tag{124}
$$

Thus we can see that Γ is not time independent, it depends on the number density of the particle and the velocity of particle, which can change with time.

5.3 Freeze-in and Freeze-out

With this Boltzmann equation, we can now understand the process of **freeze-in** and **freeze-out** of particles in the early universe. Note that from the Boltzman equation, the number of particles changes due to two effects:

- **Expansion of the universe**: which is given by the term $3H(t)N_i(t)$.
- **Interactions of the particle**: which is given by the term $-\Gamma \times \left(N_i(t) - N_i^{\text{eq}}\right)$.

Both terms are given by a rate Γ and H. We can find that both rates are **Temperature Dependent**:

- Γ depends on v the relative velocity of the particle, which is related to the temperature of the universe.
- H depends on time as univers expands, which is also related to the temperature of the universe Equation 96.

Thus, as the universe cools down, different terms will dominate at different temperature, which gives us the process of freeze-in and freeze-out.

5.3.1 Assymptotic Behavior of the Boltzmann Equation

We discuss first the asymptotic solution of the Boltzmann equation. We have two limits:

- $\Gamma(T) \gg H(T)$: in this limit, the interaction term dominates, thus we have

$$N_i(t) \approx N_i^{\text{eq}} \tag{125}$$

which means that the particle is in thermal equilibrium with the universe.

- $\Gamma(T) \ll H(T)$: in this limit, the expansion term dominates, Boltzmann equation becomes:

$$\frac{d}{dt}\left(a^3 N_i\right) = 0 \tag{126}$$

Thus, the number density deviate from the equilibrium value, and we have:

$$N(T) \approx N^{\text{eq}}(T_*)\left(\frac{a(T_*)}{a(T)}\right)^3 \tag{127}$$

where T_* is the temperature at which $\Gamma(T_*) \approx H(T_*)$. This means the time we should not only consider the expansion of the universe, but also the interaction of the particle.

The critical temperature T_* is called the **freeze-out/in temperature** of the particle, depends on $\frac{\Gamma}{H}$ is decresing or increasing with temperature, we can have freeze-out or freeze-in process.

5.4 Electron-Positron Gas Freeze-out

We finally can come to a concrete example of the freeze-out process, which is the freeze-out of the electron-positron gas in the early universe. We consider the following scattering process:

$$e^+ e^- \leftrightarrow \gamma\gamma \tag{128}$$

With this process occuring we assume that the universe is **radiation dominant** !!

5.4.1 Scattering Cross Section and Rate of Reaction

The scattering cross section of this process is given by:

$$\sigma = \begin{cases} \frac{1}{2v}\pi r_e^2 & v \ll 1 \\ \\ \frac{m_e^2}{E^2}\pi r_e^2\left(\log\left(\frac{(2E)^2}{m_e^2}\right) - 1\right) & v \approx 1 \end{cases} \tag{129}$$

where r_e is $\frac{\alpha}{m_e}$, E is the center of mass energy of the collition, and v is the relative velocity of the particle.

This cross section is depending on the velocity of the particle, which is related to the temperature of the universe. In both limits we can calculate the rate of reaction Γ:

- **Relativistic Limit**: $T \geq m_e$ in this limit we take:

$$\sigma \approx \frac{m_e^2}{E^2}\pi r_e^2\left(\log\left(\frac{(2E)^2}{m_e^2}\right) - 1\right) \approx \frac{\alpha^2}{T^2} \tag{130}$$

> **Remark**
>
> In this approximation we take $E \approx T$ for relativistic particle and we assume the logerithm term is order of 1.

and remember that the number density of relativistic particle for equilibrium case is given by:

$$N_i^{\text{eq}} = \frac{3}{4}\left(\frac{\zeta(3)}{\pi^2}\right)g_i T^3 \tag{131}$$

Thus we have:

$$\Gamma = \langle N v \sigma \rangle \sim \alpha^2 T \tag{132}$$

- **Non-Relativistic Limit**: $T \leq m_e$ in this limit we take:

$$\sigma \approx \frac{1}{2v}\pi r_e^2 \tag{133}$$

and remember that the number density of non-relativistic particle for equilibrium case is given by:

$$N_i^{\text{eq}} = g_i\left(\frac{m_i T}{2\pi}\right)^{\frac{3}{2}}\exp\left(\frac{\mu_i - m_i}{T}\right) \tag{134}$$

Thus we have:

$$\Gamma = \langle N v \sigma \rangle \sim r_e^2 (T m_e)^{\frac{3}{2}}\exp\left(\frac{-m_e}{T}\right) \tag{135}$$

> **Remark**
>
> We here take the chemical potential μ_i to be zero, which is a good approximation in the radiation dominant universe.

5.4.2 Freeze-in/out Temperature

We list out the rate of reaction and the Hubble parameter:

$$\Gamma = \begin{cases} \sim \alpha^2 T & T \geq m_e \\ \sim r_e^2 (T m_e)^{\frac{3}{2}}\exp(\frac{-m_e}{T}) & T \leq m_e \end{cases} \tag{136}$$

while the Hubble parameter (in a radiation dominant univers) is given by Equation 94, we copy it here:

$$H = \frac{T^2}{M_0} \tag{137}$$

> **Remark**
>
> We here are assuming that the universe happening all those processes is radiation dominant, which is a good approximation for the early universe. Thus, we can negelact the chemical potential of the particle, and use this Hubble parameter to calculate the freeze-out/in temperature.

We draw the plot of Γ and H as a function of temperature:

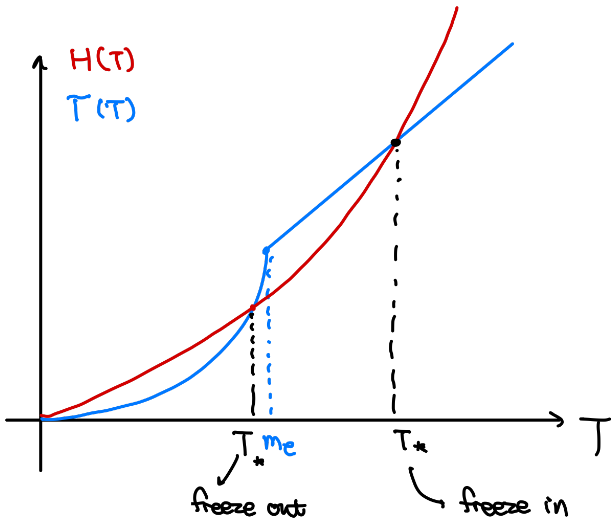

Figure 7: The plot of the rate of reaction Γ and the Hubble parameter H as a function of temperature. The intersection point of the two curves gives us the freeze-out/in temperature of the particle.

we can calculate the freeze-out/in temperature by solving the equation $\Gamma(T_*) = H(T_*)$:

- **freeze in** : this is easy to solve, we have:

$$T_{\text{in}} \sim \alpha^2 M_0 = 10^{14} \text{ GeV} \tag{138}$$

- **freeze out**: this is hard to solve, we can only get an approximate solution by taking the logerithm of the equation, we have:

$$r_e^2 (T m_e)^{\frac{3}{2}} \exp\left(\frac{-m_e}{T}\right) \sim \frac{T^2}{M_0} \tag{139}$$

which gives us:

$$T_{\text{out}} \sim \frac{m_e}{40} = 10 \text{ keV} \tag{140}$$

5.4.3 Physical Interpretation

In this process, in a radiation dominant universe, we assume that **the photon is in thermal equilibrium with the universe**, thus we have $N_\gamma \sim a^{-3}$ only changes due to the expansion of the universe. Then we can use $\frac{N_e}{N_\gamma}$ to represent the number of electrons.

- When $m_e < T < T_{\text{in}}$ the electrons are in thermal equilibrium due to the fact that $\Gamma \gg H$, while we can approximate it as a relativistic particle. Due to the fact that Equation 131, we have:

$$\frac{N_e}{N_\gamma} \sim \frac{g_e}{g_\gamma} \sim 1 \tag{141}$$

- When $T_{\text{out}} < T < m_e$ the electrons are not in thermal equilibrium due to the fact that $\Gamma \ll H$, while we can approximate it as a non-relativistic particle. Due to the fact that the number density of non-relativistic particle is given by Equation 134.

The electron number density shinks exponentially as the universe cools down.

- When $T < T_{\text{out}}$ the scatter is minor and the number of electrons is only affected by the expansion of the universe, thus now we have the number density of electrons given by as $N_e \sim N_* a^{-3}$. where N_* is the number density of electrons at the freeze-out temperature T_{out}.

If we plot the number density of electrons as a function of temperature, we can see that the number density of electrons shinks exponentially as the universe cools down as $T < m_e$, and finally freezes out at T_{out}:

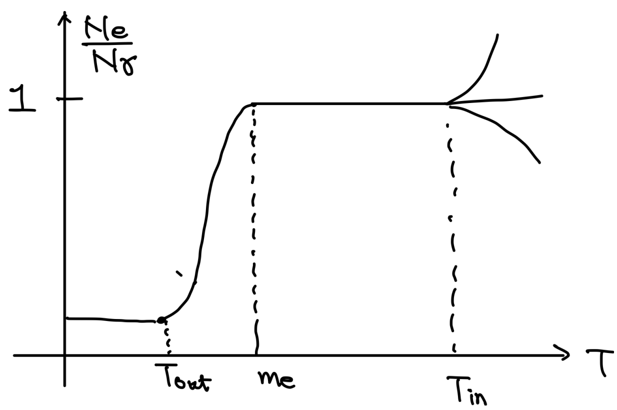

Figure 8: The plot of the number density of electrons as a function of temperature.

An important remark is that the plot is independent of the initial number density and things happening at $T > T_{\text{in}}$.

6 Lecture 6: Recombination and Neutrino Freeze Out

Now we start to focus on more concrete process happened in the universe. Now I'll list out a timeline of the process happened in the universe, which is shown in the following figure:

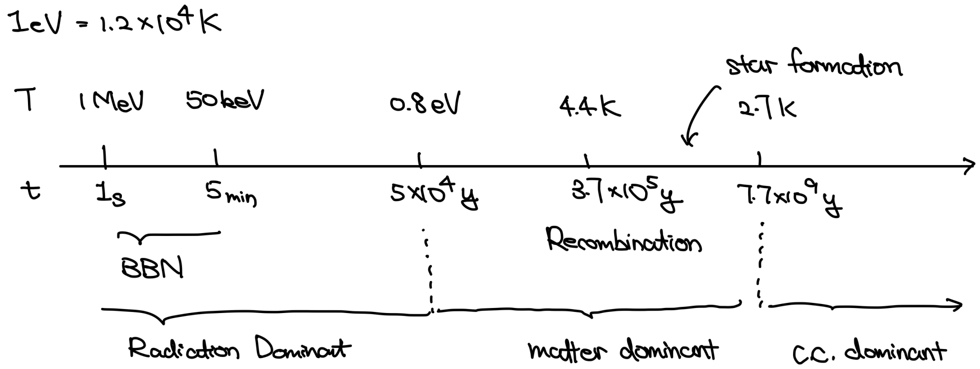

Figure 9: Timeline of the thermal history of the universe.

This lecture we will focus on the process of recombination and neutrino freeze out which happens long before, even before the BBN process.

Note that the freeze out of electrons and positrons we discussed last lecture happens right after BBN.

6.1 Recombination

This this the process when the CMB is emitted. Before the recombination, the universe is filled with plasma, which is opaque to photons, with the main reaction happening as:

$$e^- + p^+ \to H + \gamma \tag{142}$$

Yet as the universe expands and cools down, the reaction will be more difficult to happen when the temperature drops and energy is lower the binding energy of the hydrogen atom. Then the universe becomes transparent to photons, and photons are decoupled from matter, and free to propagate in the universe.

The slice when the last scattering happens is called the last scattering surface, which is the source of the CMB we observed today.

6.1.1 Saha Equation

We now focus on the process of recombination, which protons and electrons combine to form neutral hydrogen atoms. In order to calculate the recombination temperature, we need to get the number density of electrons. This can be done with the help of the Saha equation.

There exist a simple equation to describe the process of:

$$e^- + p^+ \to H + \gamma \tag{143}$$

when the elements are in thermal equilibrium, which is called the Saha equation:

Theorem 6.1 **The Saha Equation**

The Saha equation describes the ionization state of a gas in thermal equilibrium, relating the number densities of ions and electrons to the temperature and ionization energy. It is given by:

$$\frac{N_e N_p}{N_H} = \left(\frac{m_e T}{2\pi}\right)^{\frac{3}{2}} \exp\left(-\frac{I}{T}\right) \tag{144}$$

where N_i is the number density of species i, m_e is the electron mass, T is the temperature, and $I = m_e + m_p - m_H$ is the ionization energy of hydrogen.

All elements other than photons can be view as thermal equilibrium non-relativistic particles, and the equilibrium number density can be calculated as Equation 89. we copy it here for convenience:

$$N_i = g_i \left(\frac{m_i T}{2\pi}\right)^{\frac{3}{2}} \exp\left(\frac{\mu_i - m_i}{T}\right) \tag{145}$$

the polarization degree of freedom for proton and electron are both 2, and for hydrogen atom is 4, thus we have $g_e = g_p = 2$ and $g_H = 4$. Moreover, in chemical equilibrium the chemical potential of the reaction is zero, thus we have

$$\mu_e + \mu_p = \mu_H \tag{146}$$

With these preparation, we can derive the Saha equation as above.

> **Remark**
>
> Note that here we have assumed that the hydrogen mass is nearly the mass of proton.

6.1.2 Recombination Temperature

To calculate the recombination temperature, generally we have to consider a complicated scattering between photons and protons and electrons. However, to make it simple, we can only consider the process that photon can scatter with electrons, and ignore the scattering between photons and protons, which is a good approximation since the cross section of photon-electron scattering is much larger than that of photon-proton scattering.

Then we only have to focus on the **Thompson scattering** process, which is the scattering between photons and electrons:

$$\gamma + e^- \to \gamma + e^- \tag{147}$$

To do the calculation, we need some facts and assumptions:

- Assume the universe is neutral, and positrons are mostly annihilated after the freeze out of electrons and positrons, thus we have $N_e = N_p$.
- Fact: the baryon to photon ratio is $\eta = N_b / N_\gamma = 6 \times 10^{-10}$, where N_b is the number density of baryons, and N_γ is the number density of photons.
- Fact: the fraction of protons in baryons is about 0.75, thus we have $N_p = 0.75 N_b$.
- Assume that at the temperature of interest most protons have combined with electrons to form hydrogen atoms, thus we have $N_H = N_{p\ \text{original}}$.

With these assumptions, we now can see that the number density of electrons can be expressed through Saha equation assuming $N_e = N_p$ as:

$$N_e = N_H^{1/2}\left(\frac{m_e T}{2\pi}\right)^{3/4} \exp\left(-\frac{I}{2T}\right) \tag{148}$$

While N_H can be expressed through the baryon to photon ratio as:

$$N_H = N_{p\ \text{original}} = 0.75 N_b = 0.75\eta N_\gamma = 0.75\eta\left(2\frac{\zeta(3)}{\pi^2}\right)T^3 \tag{149}$$

In the final equation we use the equilibrium number density of photons, given by Equation 81. With the number density of electrons, we can calculate the Γ of Thompson scattering as:

$$\Gamma = \langle \sigma_{\gamma e} N_e v \rangle \tag{150}$$

the cross section of Thompson scattering is given by:

$$\sigma_{\gamma e} = \frac{8\pi\alpha^2}{3m_e^2} \tag{151}$$

The recombination happens when the scattering effect is as the same order as the expansion effect (freeze out), thus we have the condition:

$$H = \Gamma \quad \Rightarrow \quad B\frac{T^{9/4}}{m_e^{5/4}} e^{-I/2T} = \frac{T^2}{M_0} \tag{152}$$

where B is a constant given by:

$$B = \frac{8\pi\alpha^2}{3}\eta^{1/2}\left(\frac{2\zeta(3)}{\pi^2}\right)^{1/2}\left(\frac{1}{2\pi}\right)^{3/4} \tag{153}$$

Note that here we take $v = 1$, which is the speed of light, since the photons is involved.

We finally get the recombination temperature as:

$$T_{\text{dec}} = 0.25 \text{ eV} \sim 3000K \tag{154}$$

6.1.3 CMB Temperature

This temperature is the temperature of the photons at the time of recombination, and then afterwards the photons are decoupled from matter and free to propagate in the universe, which becomes the CMB. As the CMB propagates it can be described as free particles and has a effective temperature, which is the CMB temperature.

Today, the CMB temperature is measured to be:

$$T_{\text{CMB}} = T_{\gamma,0} = 2.725 \text{ K} \tag{155}$$

By definition of effective temperature Section 5.1.3, we have the following relation:

$$T_{\gamma,0} = T_{\text{dec}}\left(\frac{a_{\text{dec}}}{a_0}\right) \tag{156}$$

One can calculate the ratio between this temperature at recombination and the temperature of CMB today, which gives us the redshift of the recombination. remember Equation 97, we have:

$$z_{\text{rec}} \sim \frac{a_0}{a_{\text{dec}}} = \frac{T_{\text{dec}}}{T_{\text{CMB}}} \sim 1100 \tag{157}$$

which is quite at the same order as the redshift of matter-radiation equality, which is given by $z_{\text{eq}} \sim 3000$. This means that the recombination happens quite close to the matter-radiation equality.

> **Remark**
>
> By definition, the redshift should be $z = a_0/a - 1$, we just ignore the -1 since the redshift is quite large.

6.2 Neutrino Freeze Out

Now we focus on another process happend before recombination, which is the neutrino freeze out. Neutrinos interacts with electrons:

$$\nu + e^- \to \nu + e^- \tag{158}$$

And when the temperature drops, the interaction will be more difficult to happen, it will be less dominant than the expansion, and eventually the neutrinos will freeze out and decouple from the matter.

Yet this interaction is much weaker than the Thompson scattering. Thus, it is expected to stop happening before the recombination happens. Lets now calculate the freeze out temperature of neutrinos.

6.2.1 Freeze Out Temperature of Neutrinos

The cross section of the neutrino-electron scattering is given by:

$$\langle \sigma_w v \rangle \sim G_F^2 E^2 \quad G_F = g^2/M_w^2 \sim 10^{-5}\text{GeV}^{-2} \tag{159}$$

> **Remark**
>
> Note we assume that the neutrinos are massless, relativistic particles

Then we can calculate the Γ of neutrino-electron scattering as:

$$\Gamma = \langle \sigma_w v N_\nu \rangle \sim G_F^2 T^2 T^3 \tag{160}$$

> **Remark**
>
> Due to the fact we still view neutrinos as relativistic particles, we can calculate the energy of neutrinos as $E \sim T$, so we have T^2.
>
> Moreover, the number density of neutrinos can be calculated from the equilibrium number density of relativistic particles, which is given by Equation 81, thus we have $N_\nu \sim T^3$.

With this data calculated, we can calculate the freeze out temperature of neutrinos by comparing the scattering effect with the expansion effect:

$$\Gamma = H \quad \Rightarrow \quad G_F^2 T^5 = \frac{T^2}{M_0} \tag{161}$$

This gives us the freeze out temperature of neutrinos as:

$$T_*^\nu = \left(G_F^2 M_0\right)^{-1/3} \sim 2 \text{ MeV} \tag{162}$$

6.2.2 Neutrino and Photon Effective Temperature Ratio

After freeze out, the particle is not interacting with others, thus its distribution evloves as a free particle. This means that we can use the effective temperature to describe the distribution of the particle, see Section 5.1.3 the definition of effective temperature for massless particles. By definition:

$$T_{\nu,0} = T^{\nu}_{*} \left(\frac{a_*}{a_0} \right) \tag{163}$$

where $T_{\nu,0}$ is the effective temperature of neutrinos today, T_{ν_*} is the freeze out temperature of neutrinos, a_* is the scale factor at the freeze out time, and a_0 is the scale factor today.

As I have claimed in former remarks, we often roughly assume that the effective temperature is the same as the assumed equilibrium (of course its not) temperature of the universe.

$$T_{\nu,0} \sim T_{\gamma,0}(\text{aka } T_{\text{CMB}}) \sim T_0 \tag{164}$$

Yet here we meet the first example that this approximation is not good.

The effective temperature of neutrino is not the same as the CMB temperature experimentally:

$$T_{\nu,0} \neq T_{\text{CMB}} \tag{165}$$

This is due to the effect that after the neutrino freeze out, the photons are still interacting with electrons and by then positrons and electrons annihilate each other, which gives out energy to photons and heat the photon up. Here is a time line of the process:

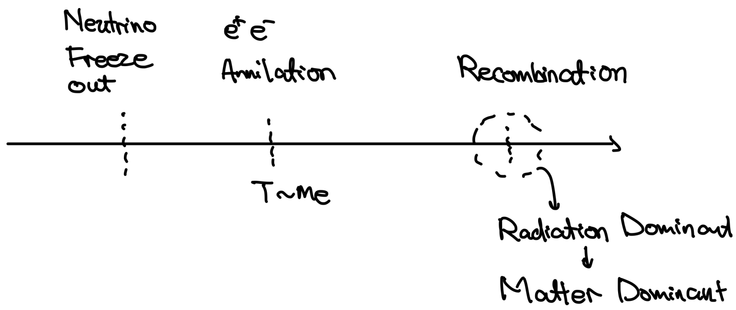

Figure 10: Timeline of the neutrino freeze out and recombination process.

We can calculate the ratio using the conservation of entropy. We think that the universe is a closed system, there's nothing "outside" the universe. Thus, the total entropy of the universe is conserved.

We further assume that the universe is radiation dominated (as we always did) and the entropy density of relativistic particle is given by Equation 87, then the total entropy of the universe is given by:

$$S = sV \sim g_* T^3 a^3 = \text{constant} \tag{166}$$

> **Remark**
>
> Here we have a a^3 due to the fact that the volume of the universe is given by $V \sim a^3$.

Then we consider right after the neutrinos freeze out, the interacting particles we assume only includes electrons, positrons and photons forming a plasma, and the neutrinos are already decoupled from the plasma.

> **Note**
>
> Here we also ignore the contribution of non-relativistic particles, which has lower energy density gives lower entropy density.

Thus the total entropy of the universe is given by:

$$S \sim \left(g_\gamma + g_{e^+e^-}\right)(T_*^\nu)^3 a_*^3 \quad g_{*\gamma} = 2, \quad g_{*e^+e^-} = \frac{7}{8}(2+2) \tag{167}$$

Then neutrino effective temperature goes like $T \sim \frac{1}{a}$, thus we have:

$$(T_*^\nu) a_* = T_{\nu,0} a_0 \tag{168}$$

Then some time later the positrons and electrons annihilate each other, the entropy of the universe is given by:

$$S \sim g_{*\gamma} T^3 a^3 \tag{169}$$

And remeber that during all these process the electron photon plasma is evolving as a relativistic equilibrium system and the temperature goes like $T \sim \frac{1}{a}$. Then the photon freeze out the effective temperature still goes like $T \sim \frac{1}{a}$. Thus we have:

$$T_{\text{rec}} a_{\text{rec}} = T_{\gamma,0} a_0 \tag{170}$$

With all these ingrediants, we can calculate the ratio between the neutrino effective temperature and the photon effective temperature as:

$$\frac{T_{\gamma,0}}{T_{\nu,0}} = \frac{a_{\text{rec}} T_{\text{rec}}}{a_* T_*^\nu} = \left(\frac{g_\gamma + g_{e^+e^-}}{g_\gamma}\right)^{1/3} = \left(\frac{11}{4}\right)^{1/3} = 1.401 \tag{171}$$

From this equation we can get the effective temperature of neutrinos today from the CMB temperature:

$$T_{\nu,0} = \frac{T_{\text{CMB}}}{1.4} \sim 2\ \text{K} \tag{172}$$

6.2.3 Cosmological Bound on Neutrino Mass

We first calculate the number density of neutrinos today, which is given by the equilibrium number density of relativistic particles Equation 81:

$$N_{\nu,0} = \left(\frac{3}{4}\right) g_\nu \left(\frac{\zeta(3)}{\pi^2}\right) T_{\nu,0}^3 \tag{173}$$

There are different spicies of neutrinos, and due to the fact they interact with electrons in the same style we take this number density to be true to all spicies of neutrinos. Due to the relation between the photon effective temperature and the neutrino effective temperature, we can express the number density of neutrinos today through the photon number density today, which is given by:

$$N_{\nu,0} = \frac{3}{22} N_{\gamma,0} \tag{174}$$

In all above calculations we take the neutrinos to be massless particles, however, in reality from the neutrino oscillation experiments we know one neutrino is massless and others may have mass.

We want to use this number density to bound the mass of neutrinos. To do this, we know in fact the energy density of different spicies of particles are measureable, and for mass particles we have data;

$$\Omega_M < 0.4 \tag{175}$$

There is a bound that:

- Neutrino can not take up all matter in the universe

Mathematically, we have:

$$\Omega_\nu = \frac{\rho_{\nu,0}}{\rho_c} < \Omega_M \sim 0.4 \tag{176}$$

As non-relativistic particles, the energy density of massive neutrinos can be:

$$\rho_{\nu,0} = \sum_i m_{\nu,i} N_{\nu,0} = \sum_i m_{\nu,i} \left(\frac{3}{22}\right) N_{\gamma,0} \tag{177}$$

Eventually, we can get the bound on the mass of neutrinos as:

$$\sum_i m_{\nu,i} \leq 10 \text{ eV} \tag{178}$$

> **Remark**
>
> There are modern cosmological data that can give us a much stronger bound on the neutrino mass, which is given by:
>
> $$\sum_i m_{\nu,i} \leq 0.2 \text{ eV} \tag{179}$$

7 Lecture 7: Big Bang Nucleosynthesis

This is the process before the freeze out discussed in the previous lectures. During BBN the protons and neutrons combines into light nuclei, such as deuterium, helium-3, helium-4 and lithium-7.

As a result of BBN, we have the unexplained facts we used in recombination calculation of number density of hydrogen in recombination:

- The abundance of protons are of order of 75% and the abundance of helium-4 is of order of 25% and others are negligible.

Now lets have a look at what happened during BBN.

7.1 Neutron to Proton Ratio at the Beginning of BBN

7.1.1 Generation of Neutrons

At the beginning of BBN, the neutrons are produced and destroyed by the weak interaction process:

$$p + e^- \leftrightarrow n + \nu_e \tag{180}$$

protons and electrons can combine to produce neutrons and electron neutrinos.

> **Remark**
>
> According to the time line, in fact neutrinos are already freeze out. However, freeze out doesn't really means to decouple, its just the interaction rate is smaller than the Hubble expansion rate, but it doesn't mean that there is no interaction at all.

One can calculate the freeze out temperature of the above process:

$$T^* \sim T_\nu^* \sim 2 \text{ MeV} \tag{181}$$

Thus it happens quite the same time as neutrino freeze out. A more precise calculation gives the freeze out temperature of the above process to be around:

$$T^* \sim 0.8 \text{ MeV} \tag{182}$$

which is slightly after the neutrino freeze out.

7.1.2 Saha Equation

Then we calculate the number density. Remeber in recombination we calculate the number density of electrons using Saha equation. Here we will use the same method to calculate the number density of neutrons and protons.

Remember the number density of massive particles:

$$N_A = g_A \left(\frac{m_A T}{2\pi} \right)^{3/2} \mathrm{e}^{(\mu_A - m_A)/T}, \tag{183}$$

and we assume that equilibrium holds, which means that the chemical potential of the above process should satisfy:

$$\mu_p + \mu_e = \mu_n + \mu_{\nu_e} \tag{184}$$

If we approximate the mass of protons and neutrons to be the same, then we have:

$$\frac{N_n}{N_p} = \exp\left(-\frac{m_n - m_p}{T} + \frac{\mu_e - \mu_{\nu_e}}{T}\right) \tag{185}$$

note here we use $\mu_n - \mu_p = \mu_e - \mu_{\nu_e}$ which is from the chemical potential relation. We already know that $m_n - m_p \sim 1.3$ MeV. We now try to argue that: μ_{ν_e}/T and μ_e/T are negligible.

7.1.3 Negeligible Chemical Potential

There is a basic result in thermaldynamics that the difference of number density between particle and antiparticle is given by:

Theorem 7.1 **Particle-Antiparticle Asymmetry**

The difference of number density between particle and antiparticle at $T \gg m, \mu$ is given by:

$$\Delta N_i = \begin{cases} \mu T^3/6 & \text{fermion} \\ \mu T^3/3 & \text{boson} \end{cases} \tag{186}$$

The proof is that we assume the Boltzmann distribution and take a limit. Then we use the fact that the chemical potential of antiparticle is the negative of the chemical potential of particle, which means that $\mu_{\bar{i}} = -\mu_i$.

With this result, we can see that:

$$\frac{\mu_e}{T} = 6\frac{\Delta N_e}{T^3} \tag{187}$$

if we assume the universe is charge neutral, then we have $\Delta N_e = N_p$, and we know that:

$$N_p \sim \eta N_\gamma \sim \eta T^3 \tag{188}$$

Thus we see:

$$\frac{\mu_e}{T} \sim 6\eta \ll 1 \tag{189}$$

We here just **assume** that the chemical potential of neutrinos is also negligible.

7.1.4 Neutron to Proton Ratio

With above preparation, we can see that from the Saha equation, we have:

$$\frac{N_n}{N_p} = \exp\left(-\frac{m_n - m_p}{T}\right) \tag{190}$$

We plug in the freeze out temperature $T^* \sim 0.8$ MeV and the mass difference $m_n - m_p \sim 1.3$ MeV, we can get the neutron to proton ratio at the beginning of BBN:

$$\frac{N_n}{N_p} \sim \frac{1}{5} \tag{191}$$

Remark

There is another effect that we haven't considered, which is the neutron decay. However, one can see that the above process happened quite fast $\sim 1s$ which is much smaller than the neutron decay time $\sim 880s$, thus we can ignore the effect of neutron decay at the beginning of BBN.

7.2 Nuclear Reaction I: Start of Nuclear Reaction

After we have neutrons and protons fixed, they can combine into light nuclei.

7.2.1 General Picture

Here is a general picture of the chain of nuclear reaction during BBN:

$$\begin{gathered} p+n \leftrightarrow D+\gamma \\ D+D \leftrightarrow {}^3\mathrm{H}+p, \quad D+D \leftrightarrow {}^3\mathrm{He}+n \\ D+{}^3\mathrm{He} \leftrightarrow {}^4\mathrm{He}+\gamma \\ {}^3\mathrm{He}+n \leftrightarrow {}^4\mathrm{H}+p \\ \ldots \end{gathered} \tag{192}$$

The reaction happens so on and so on. One can naively think that the neuclei with the largest binding energy per nucleon should be the most abundant, for it takes more energy to break every nucleon out of the nucleus. The binding energy is given by:

$$I_A = Zm_p + (A-Z)m_n - m_A \tag{193}$$

and the binding energy per nucleon is given by:

$$\frac{I_A}{A} = \frac{Zm_p + (A-Z)m_n - m_A}{A} \tag{194}$$

Here is a table of the binding energy per nucleon for light nuclei:

Nucleus	Z	A	I/A (MeV)
^{2}H (deuteron)	1	2	1.11
^{3}H (triton)	1	3	2.83
^{3}He	2	3	2.57
^{4}He	2	4	7.07
^{6}Li	3	6	5.33
^{7}Li	3	7	5.61
^{9}Be	4	9	6.46
^{11}B	5	11	6.93
^{12}C	6	12	7.68
^{16}O	8	16	7.98

We can see that the binding energy per nucleon of helium-4 is a local maximum. Thus, we may expect in the limit of time for reaction (due to universe is expanding and cooling down), the most abundant nucleus should be helium-4, which is consistent with the observation.

And time is not enough for the reaction to produce heavier nuclei, thus we don't have C, O, etc. in that stage.

7.2.2 Producing of Deuterium

To produce all other nuclei, we need to first produce deuterium. The reaction is:

$$p+n \leftrightarrow D+\gamma \tag{195}$$

However, in the begining BBN as the neutrons just freeze out, the temperature is too high, that even if the deuterium is produced, it will be immediately destroyed by the high energy photons. The reaction of creating deuterium can be viewed as not happening.

As temperature goes lower, the deuterium can survive. Thus, here we want to calculate the temperature at which deuterium can survive and being generated, this will be the beginning of neuclear reaction.

To do this we need assumptions:

1. We assume equilibrium holds. This is the same as we think as long as the reaction happens (deuterium survives), it happens fast enough to maintain equilibrium.
2. We assume only this nuclei reaction happens and turn off the others.

With equilibrium assumed, we can use the Saha equation to calculate the number density of deuterium, and we take the temperature that :

$$\frac{N_D}{N_n} \sim 1 \tag{196}$$

as the temperature at which deuterium can survive and being generated.

The Saha equation for the above reaction is:

$$\frac{N_n N_p}{N_D} = \left(\frac{m_p T}{2\pi}\right)^{3/2} \mathrm{e}^{I_D/T} \tag{197}$$

We can use the relation that $N_p = \eta N_\gamma$ and $N_\gamma \sim T^3$, and I_D is the binding energy of deuterium $I_D \sim 2.23$ MeV. Thus one can calculate the temperature at which deuterium can survive and being generated:

$$\frac{N_n}{N_D} \sim 1 \quad \to \quad T_{\mathrm{NS}} \sim 70 \text{ keV} \tag{198}$$

we can see this happens much later than the freeze out of neutron to proton conversion, which is around 0.8 MeV.

Notice that we assume that the neutrons have already freezed out. However, in fact during the time after neutrons freeze out and deuterium can survive, the neutrons may dacay. The actual neutron to proton ratio at the time of deuterium survival is around $1/7$ instead of $1/5$.

7.3 Nuclear Reaction II: Abundance of Nuclei

7.3.1 Helium-4 Abundance (Rough Estimation)

After deuterium is produced, the reaction can go on and on. As we analyzed before, the most abundant nucleus should be helium-4. We can make a rough estimation that all neutrons are used to produce helium-4, thus determines the abundance of helium-4.

First we have to define a quantity that capture the abundance of a nucleus:

Definition 7.2 **Abundance of Nucleus**

The abundance of a nucleus is defined as :

$$x_A = \frac{AN_A}{N_b} \qquad \sum_A x_A = 1 \tag{199}$$

where N_b is the number density of baryons, and A is number of baryons in the nucleus, and N_A is the number density of the nucleus.

This means how many percentage of baryons come and form a certain nucleus.

Then if we assume all neutrons are used to produce helium-4, then we have:

$$x_{\text{He}} = \frac{4N_{\text{He}}}{N_b} = \frac{4N_n/2}{N_b} \tag{200}$$

we already know that $\frac{N_n}{N_p} \sim 1/7$ at the time of deuterium survival, and also by definition $N_p + N_n = N_b$, thus we can calculate the abundance of helium-4:

$$x_{\text{He}} \sim \frac{4}{2}\frac{1/7}{1+1/7} \sim 0.25 \tag{201}$$

This gives an answer to the result used in recombination that the abundance of helium-4 is around 25% and the abundance of protons is around 75%.

7.3.2 Saha Equation for Nuclear Reaction

We want to make a more precise calculation of the abundance for all nuclei. To do this, we still make assumptions of equilibirum for all reactions. And view the reaction as a whole, we can write the reaction as:

$$Zp + (A-Z)n \leftrightarrow {}^{A}\text{Nucleus} + \gamma \tag{202}$$

Then we use can derive a Saha equation for the reaction. Equilibrium means that the chemical potential of the reaction should satisfy:

$$Z\mu_p + (A-Z)\mu_n = \mu_A \tag{203}$$

this gives us a Saha equation for the reaction:

Theorem 7.3 **Saha Equation for Nuclear Reaction**

The Saha equation for the nuclear reaction is given by:

$$N_A = N_p^Z N_n^{A-Z} 2^{-A} g_A A^{3/2} \left(\frac{2\pi}{m_p T}\right)^{\frac{3}{2}(A-1)} e^{I_A/T}, \tag{204}$$

where I_A is the binding energy of the nucleus, and g_A is the degeneracy of the nucleus.

$$I_A = Zm_p + (A-Z)m_n - m_A \tag{205}$$

and in the non-exponential part, we assume that $m_A = Am_p$.

This equation can be written in terms of abundance:

$$x_A = x_p^Z x_n^{A-Z} N_b^{A-1} 2^{-A} g_A A^{5/2} \left(\frac{2\pi}{m_p T}\right)^{\frac{3}{2}(A-1)} e^{I_A/T}. \tag{206}$$

and we know that the number density of baryons is given by:

$$N_b = \eta N_\gamma = \eta \left(2\frac{\zeta(3)}{\pi^2}\right) T^3 \tag{207}$$

we now do an apporximation that the number density of protons are quite the number density of baryons, $x_p \sim 1$ then we can have a relation between number density of He and the number density of neutrons, if we inverse the identity we have:

$$x_n = x_{\text{He}}^{1/2} \eta^{3/2} \left(\frac{T}{m_p}\right)^{-9/4} e^{-I_{\text{He}}/(2T)} \tag{208}$$

Then we make a second approximation that the number density of He is quite the number density of baryons, $x_{\text{He}} \sim 1$, then we can have the number density of neutrons.

Now we have the ingrediant of N_B, x_n, x_p we can calculate the abundance of any particles at certain temperature, the final relation is

Theorem 7.4 **Abundance of Nucleus**

The abundance of a nucleus is given by:

$$\begin{aligned} x_A &\sim \left[\eta \left(\frac{T}{m_p}\right)^{3/2}\right]^{3/2Z-1/2A-1} \exp\left(I_A/T - \left(\frac{Z-A}{2}\right) I_{\text{He}}/T\right) \\ &\sim \exp\left(\frac{I_A}{A-Z}/T - \frac{I_{\text{He}}}{2}/T\right) \end{aligned} \tag{209}$$

This relation is mainly based on that He dominant.

Here we can see that the abundance is highly depend of the binding energy per neutron:

$$\frac{I_A}{A-Z} \tag{210}$$

This also justifies why He dominant for it has the largest binding energy per neutron.

7.4 Abundance of Deuterium

Using this formula, we can calculate the abundance of deuterium, helium-3, lithium-7, etc. at the temperature of T_{NS} which gives us:

$$x_D \sim 10^{-120} \tag{211}$$

however, in reality the abundance of deuterium is around 10^{-5}. This is because the assumption of equilibrium doesn't hold for deuterium, they freeze out before they are all used to produce helium-4.

Consider the reactions:

$$\begin{gathered} p + n \leftrightarrow D + \gamma \\ D + D \leftrightarrow {}^3\text{H} + p, \quad D + D \leftrightarrow {}^3\text{He} + n \end{gathered} \tag{212}$$

we now write out the Boltzmann equation for deuterium:

$$\frac{dN_D}{dt} + 3HN_D = -\langle \sigma v N_D \rangle N_D \tag{213}$$

we now calculate the abundance of deuterium at the time of freeze out, if we take $N_p \sim N_b$ then we have:

$$\begin{gathered} N_p \sim \eta N_\gamma \sim \eta T^3 \\ H \sim \frac{T^2}{M_0} \sim \Gamma = \sigma v N_D \end{gathered} \tag{214}$$

Thus we have:

$$\frac{N_D}{N_p} \sim \frac{1}{\sigma v M_0 \eta T} \tag{215}$$

we can plug in some numbers:

$$T \sim T_{\mathrm{NS}} \quad \sigma v \sim 10^{-17} \frac{\mathrm{cm}^3}{s} \quad M_0 \sim 10^{18}\ \mathrm{GeV} \quad \eta \sim 10^{-9} \tag{216}$$

Finally we can get the abundance of deuterium at the time of freeze out:

$$x_D \sim 10^{-5} \tag{217}$$

Notice that this number is quite sensitive to the value of η, thus we can use the abundance of deuterium to constrain the value of η.

8 Lecture 8: Baryongenesis and Dark Matter

Before the BBN process, we need to first have a lot of baryons in the universe. In cosmology we initially assume that:

- There are equal number of baryons and anti-baryons in the early universe.

> **Note**
>
> Note that this is different from the electron and positron case, where we assume that electric charge is conserved, thus we have more electrons than positrons in the early universe, to be equal to the number of protons, thus the universe is electrically neutral.

Thus it is clear that in the early universe, there should be a process of:

1. Bayongenesis: the number of baryons and anti-baryons are not equal, there exist a:

$$\frac{\Delta n}{n} \ll 1 \tag{218}$$

2. Baryons and anti-baryons annihilate each other, and only a small fraction of baryons survive after the annihilation process.

This process is currently not well tested, however, we can still talk about some of the theoretical models of baryogenesis.

8.1 General Picture

8.1.1 General Picture of Baryongenesis

A modern assumed picture of the process is as follows:

- At $T \gg 1$ GeV we have baryon symmetry:

$$n_B \sim n_{\overline{B}} \sim n_\gamma \sim T^3 \tag{219}$$

- Then some process of baryogenesis happens, that makes:

$$\frac{\Delta n}{n_B} \sim \eta_B \sim 10^{-10} \tag{220}$$

and now the asymetry is fixed and baryon number $\Delta n = n_B - n_{\overline{B}}$.

- Finally, the baryons and anti-baryons annihilate each other, and only a small fraction of baryons survive after the annihilation process.

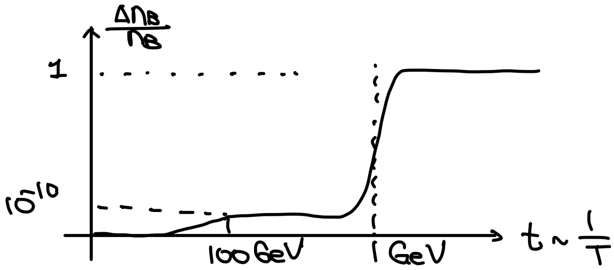

Figure 11: A picture of the process of baryogenesis.

8.1.2 Sakharov's Conditions

There are several conditions for baryogenesis to happen, which are called the Sakharov's conditions:

1. **Baryon number violation**: there must be some process that violates the baryon number, making $B = n_B - n_{\overline{B}}$ unconserved, otherwise we cannot generate a baryon asymmetry.
2. **C and CP violation**: there must be some process that violates the C and CP symmetries.
3. **Departure from thermal equilibrium**: there must be some process that takes place out of thermal equilibrium, otherwise the baryon asymmetry will be washed out by the inverse process.

We then can see that the **standard model** is a good candidate for baryogenesis, since it has all the three conditions.

8.2 Baryongenesis Process

A fact is that the **standard model** statisfies all three Sakharov's conditions, thus it is a good candidate for baryogenesis.

8.2.1 Sphaleron Process

Naively, the standard model preserved baryon number. In reality we didn't see any protons decay.

- Indeed, perturbatively, the standard model preserved baryon number
- However, there are non-perturbative processes that violate baryon number, which are called the **sphaleron** processes.

The decay rate of the sphaleron process is given by:

$$\Gamma_{\text{sph}} \sim e^{-E_{\text{sph}}/T} \quad E_{\text{sph}} \sim 100 \text{ GeV} \tag{221}$$

we can see that at low temperture, the sphaleron process is highly suppressed, however, at high temperture, the sphaleron process can be very efficient.

Naively, we see a process of violating baryon number. However, if we look deeper, it has the opposite effect we want. The sphaleron process is of thermal equilibrium if $T \geq 100\text{GeV}$. Thus, if we generate a baryon asymetry throught whatever process in early universe, the sphaleron process will wash out the baryon asymmetry.

- However, in SM $B - L$ is an exact symmetry, thus if some process generates a non-zero $B - L$ asymmetry, then the sphaleron process will not wash out the baryon asymmetry.

> **Remark**
>
> Note that B here is baryon number $B = n_B - n_{\overline{B}}$ and L is lepton number $L = n_L - n_{\overline{L}}$.

Thus a crutial point of having baryongenesis to survive the sphaleron process is to have a non-zero $B - L$ asymmetry. There are several models of baryogenesis

1. **Beyond SM**: A common assumption is to add heavy particles in SM that can decay in a way that violates $B - L$ symmetry, thus generating a non-zero $B - L$ asymmetry.
2. **Within SM**: there is a possibility of generating a non-zero $B - L$ asymmetry within the standard model, which is called the **electroweak baryogenesis**.

> **Remark**
>
> However, this is based on some assumptions that didn't really hold in reality, thus it is not a very good model of baryogenesis.

8.2.2 Electroweak Baryogenesis

This mechanism is based on a "wrong" assumption that the electroweak phase transition is a first order phase transition, which is not the case in reality. This subsection will just be a sketch and details will be missing.

If we assume the electroweak phase transition is a first order phase transition. Then as the phase transition happens, there will be two local minima of the free energy coexisting, and there will be bubbles of the true vacuum.

> **Note**
>
> Just imagine as the boiling water, there are bubbles of steam coexisting with the liquid water, and the steam is the true vacuum and the liquid water is the false vacuum, yet a local minima.

Now sphaleron processes at the bubble wall will violate the baryon number, and create more baryons in the bubbles. And as we all end up in the true vacuum, the sphaleron process will stop as the temperature drops, thus the baryon asymmetry generated in the bubbles will be preserved.

8.3 Residue of Antibaryons

We now try to calculate how much of the anti-baryons can survive after all the annihilation process. See page 316 of [1] for details.

Before calculating the freeze out temperature, we compare Γ with H. Now we use a different but equivalent way of doing this. Remember the Boltzmann equation:

$$\frac{dN_X}{dt} + 3HN_X = -\langle\sigma_{\text{ann}} \cdot v\rangle \cdot (N_X^2 - N_X^{eq2}). \tag{222}$$

We ask when the annihilation process freezes out, we see that:

- The expansion of universe make the equilibrium number density N^{eq} decrease.
- To maintain the equilibrium, the annihilation process need to be efficient enough to follow this decrease, and the characteristic speed of this process is given by:

$$-\langle\sigma_{\text{ann}} v\rangle \cdot N^{eq2} \tag{223}$$

Thus we now have a different way of comparing Γ with H, which is to compare:

$$\langle\sigma_{\text{ann}} v\rangle \cdot N^{eq2} \quad \text{and} \quad \frac{d}{dt}N^{eq} \tag{224}$$

The freeze out temperature is then given by the condition:

$$\langle\sigma_{\text{ann}} v\rangle \cdot N^{eq2} \sim \frac{d}{dt}N^{eq} \tag{225}$$

Now we generalize this to the case of baryons and anti-baryons. We have the Boltzmann equation:

$$\frac{d(N_B a^3)}{dt} = -\langle\sigma_{\text{ann}} v\rangle \cdot \left(N_B N_{\overline{B}} a^3 - N_B^{eq} N_{\overline{B}}^{eq} a^3\right) \tag{226}$$

Thus the freeze out temperature is given by the condition:

$$\left|\frac{dN^{eq}_{\overline{B}}a^3}{dt}\right| \sim \langle\sigma_{\text{ann}}v\rangle \cdot N^{eq}_B N^{eq}_{\overline{B}} a^3. \tag{227}$$

> **Remark**
>
> We use this freeze out condition is because this is an annhilation process the number density appears in Γ is that of the other particles.

Now we use some trick to evaluate this condition. A fact is that the particle and anti-particle have opposite chemical potential, thus we have at equilibrium:

$$N^{eq} = \left(\frac{m_p T}{2\pi}\right)^{3/2} e^{-\frac{m_p-\mu_B}{T}}, \quad N^{eq}_{\overline{B}} = \left(\frac{m_p T}{2\pi}\right)^{3/2} e^{-\frac{m_p+\mu_B}{T}}. \tag{228}$$

Thus we know that:

$$N^{eq}_B N^{eq}_{\overline{B}} = \left(\frac{m_p T}{2\pi}\right)^{3} e^{-2\frac{m_p}{T}} \tag{229}$$

Then we use a trick of plugging in the number density of baryons in terms of photons (quite a common trick for evaluation) $N_B = \eta N_\gamma$ and $N_\gamma \sim T^3$, we have:

$$N^{eq}_{\overline{B}} \sim \left(\frac{m_p^3}{\eta}\right) e^{-2\frac{m_p}{T}} \tag{230}$$

We can see that the number density of anti-baryons is exponentially suppressed by the factor $e^{-2\frac{m_p}{T}}$, this dominate the time evolution of $a^3 N^{eq}_{\overline{B}}$ on the LHS. Thus we approximate:

$$\left|\frac{dN^{eq}_{\overline{B}}a^3}{dt}\right| \sim \left|\frac{dN^{eq}_{\overline{B}}}{dt}\right| a^3 \tag{231}$$

Now remember in the radiation dominate era, we have:

$$T = \sqrt{\frac{M_0}{2t}} \text{ thus } \left|\frac{\dot{T}}{T}\right| = \frac{1}{2t} = H \tag{232}$$

Thus the LHS of the freeze out condition is given by:

$$\left|a^3 \frac{dN^{eq}_{\overline{B}}}{dt}\right| \sim a^3 \frac{m_p}{T} H \cdot N^{eq}_{\overline{B}}. \quad \text{where } H = \frac{T^2}{M_0} \tag{233}$$

Now the freeze out condition is purely a function of temperature and constants, we can solve it to get the freeze out temperature. The result is:

$$T \sim 10 \text{ keV} \tag{234}$$

We plug in this temperature into Equation 230 to get the number density of anti-baryons at freeze out, and we get:

$$N^{eq}_{\overline{B}} \sim 10^{-10^5} \tag{235}$$

which is an extremely small number, thus we can safely say that there is no anti-baryons left after the annihilation process.

9 Lecture 9: Dark Matters

This lecture is about the dark matter.

9.1 I: Evidence for Dark Matter

We first have to convince that there do exist dark matters. The evidence for dark matter is quite strong, and we have multiple lines of evidence.

1. **FRW Evolution**: remember the FRW Metric evolve depends on the energy density of the universe. We can measure and calculate the energy density of matter in the universe, and we can measure the energy density of Baryons by observation.

The difference between the total matter density and the baryon density tells us something is missing, and that is the dark matter.

2. **Galaxy Rotation Curves**:

If we assume classical mechanics, and make the assumption that the galaxy is just masses going around. The baryons are localized as observation:

$$I_B \sim I_0 \exp\left(-\frac{r}{r_0}\right) \tag{236}$$

Thus we can just assume a large body in the middle and the rest of the mass is just a point mass. Then we can calculate the rotation speed distribution:

$$v \sim \sqrt{\frac{1}{r}} \tag{237}$$

However, the observation shows that the rotation speed is almost constant:

$$v \sim \text{const} \tag{238}$$

which means there must be some mass distribution that is not localized in the center:

$$M(r) \sim r \Rightarrow \rho_{\text{DM}} \sim \frac{1}{r^2} \tag{239}$$

Modern more detailed observation now shows that in fact the dark matter density is not exactly 1/r^2, but it is close to that:

$$\rho_{\text{DM}} \sim \frac{r_c^2}{r^2 + r_c^2} \tag{240}$$

3. **Gravitational Lensing**: We can look at massive galaxies and estimate the mass by looking at the lensing effect.

4. **Bullet Cluster**: We can see that the core of baryons stuck in the middle yet the dark matter passed through.

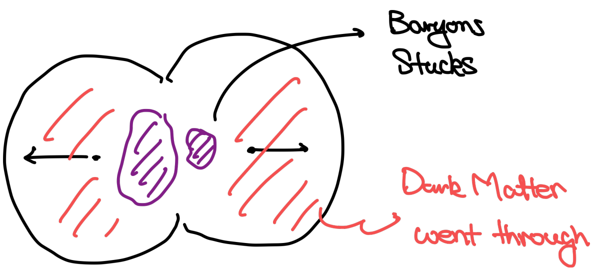

Figure 12: The bullet cluster. The purple part is the baryonic matter, and the pink part is the dark matter.

This is a very strong evidence for the existence of dark matter, and it also shows that the dark matter is not just some modification of gravity, but it is some real matter that has very weak interaction with the baryons. This also put bounds on the interaction strength of dark matter.

5. **Various Measure of BBN, CMB, and LSS**: all point to the same conclusion that there must be some dark matter in the universe.

9.2 II: Candidates for Dark Matter

A dark matter should be **stable** and **weakly interacting**. We want to know what kinds of particles can have these properties and form a good candidate for dark matter.

9.2.1 Neutrinos and Mass Bounds

- **Fermionic DM Mass Bound**

A standard model candidate for dark matter is the neutrino, cause it is only one that interacts weakly enough. However, the neutrino is not the dark matter, because it is too light.

We can calculate the number (not number density but we still use N_ν) of Neutrinos in a galaxy, due to it is a fermion, we have:

$$N_\nu \leq \frac{1}{(2\pi)^3}\int d^3p d^3x \sim p^3 r^3 \tag{241}$$

thus the total mass of the Neurtinos in the galaxy is:

$$M_\nu \sim N_\nu m_\nu \sim m_\nu^4 v^3 r^3 \tag{242}$$

We can estimate the velocity by the bound of:

$$v^2 \leq \frac{GM_\nu}{r} \tag{243}$$

This gives us a lower bound on the mass of the neutrino:

$$m_\nu > \left(\frac{1}{Gvr^2}\right)^{\frac{1}{4}} \tag{244}$$

We can plug in sum numbers from our galaxy, and we get:

$$m_\nu \geq 30 \text{ eV} \tag{245}$$

Thus for any fermionic Dark Matter, we have a mass bound from below, and apparently the neutrino is too light to be the dark matter.

- **Bosonic DM Mass Bound**

For a bosonic DM we also have a mass bound, but from another consideration. The de Broglie wavelength of the bosonic DM should be smaller than the halo radius:

$$\lambda = \frac{h}{mc} \sim \frac{1}{m} \leq r_c \tag{246}$$

This gives us a lower bound on the mass of the bosonic DM:

$$m \geq 10^{-20} \text{ eV} \tag{247}$$

We can see that this is very weak bound, and thus bosonic DM can be very light.

9.2.2 QCD Axion

Basic Axion Mass: This is originally as an explanation to the QCD strong CP problem. Moreover this also gives a good candidate for the dark matter. Due to experimental constraints, the axion mass should be around:

$$10^{-11}\ \text{eV} \leq m_a \leq 10^{-2}\ \text{eV} \tag{248}$$

If the axion mass is:

$$m_a \leq 10^{-5}\ \text{eV} \tag{249}$$

Then it serve as a good candidate for the dark matter. Axion at 0 temperature has a potential:

$$V_a|_{T=0} = -\Lambda_{\text{QCD}}^4 \cos\left(\frac{a(x,t)}{f_a}\right), \quad a(x,t) \text{ is the axion field} \tag{250}$$

at $a(x,t) = 0$ is a saddle point of this potential. Around the saddle point, we can expand the potential, which gives a mass term:

$$m_a^2 = \frac{\Lambda_{\text{QCD}}^4}{f_a^2} \tag{251}$$

we now believe this $f_a < m_p$, this gives us a lower bound on the axion mass:

$$m_a > 10^{-11}\ \text{eV} \tag{252}$$

other consideration gives us the upper bound of 10^{-2} eV.

Dark matter Candidate:Now we start to look at the behavior of Axion as temperature lowers and how it potentially becomes a dark matter. Axions can be described by a scalar field:

$$S = \int d^4x\sqrt{-g}\left[\frac{f_A^2}{2}\partial_\mu\theta\partial^\mu\theta - V(\theta)\right] \tag{253}$$

where we use $\theta = \frac{a(x,t)}{f_a}$ as the dynamical variable, and the 0 temperature potential is given as above. The equation of motion for the axion field is:

$$\ddot{\theta} + 3H\dot{\theta} + m_A^2\theta = 0 \tag{254}$$

- At high temperature, the H term dominates, thus the axion field is not occilating, but freezed at some initial value, say θ_i.
- As the temperature lowers, the H term becomes smaller, and at some point the m_a term becomes dominant, and the axion field starts to oscillate around the minimum of the potential.

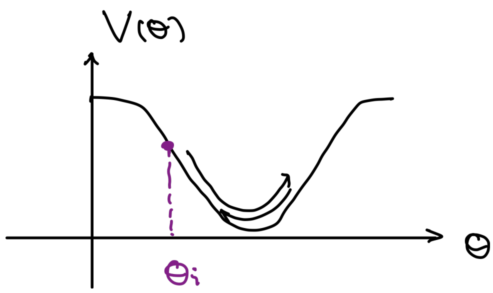

Figure 13: The axion field starts to oscillate when the m_a term becomes

We first estimate the initial energy density of the axion field when it starts to oscillate. The axion field is freezed at θ_i before it starts to oscillate, thus the initial energy density is fully given by the potential energy:

$$\rho_a \sim V(\theta_i) \sim m_a^2 f_a^2 \theta_i^2 \tag{255}$$

Then the axion field starts to oscillate, and the energy density of the axion field will dilute as matter, we can calculate the number density of the axions:

$$N_a = \frac{\rho_a}{m_a} \sim m_a f_a^2 \theta_i^2 \tag{256}$$

Now we want to calculate the number density of axions today. We know that the axion field is weak interacting, thus the number density behaves as freezed out case Equation 127 :

$$N_{a,0} = N_a \left(\frac{a_i}{a_0}\right)^3 \tag{257}$$

We know in radiation dominated era, $a \sim \frac{1}{T}$, thus we can rewrite the above equation as:

$$N_{a,0} = N_a \left(\frac{T_0}{T_i}\right)^3 \tag{258}$$

This gives us a current energy density of the axion field:

$$\rho_{a,0} = m_a N_{a,0} \sim m_a^2 f_a^2 \theta_i^2 \left(\frac{T_0}{T_i}\right)^3 \tag{259}$$

remember when the axion starts to oscillate, it satisfies the condition:

$$H(T_i) \sim m_a \tag{260}$$

In a radiation dominated era, we have:

$$T_i = (m_a M_0)^{\frac{1}{2}} \tag{261}$$

Plug this into the above equation, and use the relation of $f_a^2 = \frac{\Lambda_{\mathrm{QCD}}^4}{m_a^2}$, we finally get:

$$\rho_{a,0} \sim T_0^3 \frac{\Lambda_{QCD}^4}{M_0^{3/2} m_A^{3/2}} \theta_i^2, \tag{262}$$

For axions to be a dark matter, its energy density have to be large enough to match the observed dark matter density (or we use $\Omega_a \sim \Omega_{\mathrm{DM}}$),if we take the largest possible initial field condition $\theta_i = \pi$ this gives:

$$m_a \leq 10^{-6} \text{ eV} \tag{263}$$

9.2.3 Weakly Interacting Massive Particles (WIMPs)

There are some fermions that behave quite like neutrinos, but heavier and more stable. They are called the WIMPs.

- They are weakly interacting, and interacts with the electron-weak force.
- They are productions of common "freeze-out" mechanism, from annihilation into SM particles.

$$X + \overline{X} \to \text{SM particles} \tag{264}$$

However, we assume they cannot decay into SM particles, thus they are stable.

The freeze out happened when:

$$\Gamma_x = \langle \sigma v N_x \rangle \sim H \tag{265}$$

The particle number density is given by:

$$N_x \sim (m_x T)^{\frac{3}{2}} \exp\left(-\frac{m_x}{T}\right) \tag{266}$$

We can calculate the freeze out temperature by plugging the above equation into

$$\Gamma_x \sim H \tag{267}$$

Then the abundance from number density and radiation dominant evolution:

$$\Omega_x \sim 10^{-10} \frac{GeV^2}{\sigma_0} \log(M_0 M_x \sigma_0), \tag{268}$$

where $\sigma_0 = \sigma v$. For a electro weak procedure we can approximate σ_0. Thus gives a 1 to 1 relation between the WIMP mass and the abundance. If $\Omega_x = \Omega_{\mathrm{DM}}$ We can match the mass. However, these particles are not found yet.

9.2.4 Primordial Black Holes

10 Cosmological Perturbation Theory I

The universe is not actually homogeneous, yet have some inhomogeneities. We now investigate this. We try to do this in the linearised theory, which assume:

$$\frac{\delta\rho}{\rho} \ll 1 \tag{269}$$

On small scales this is indeed untrue, but on large scales this is still a good approximation. Notice that the inhomogeneities of the CMB is of:

$$\frac{\delta\rho}{\rho} \sim 10^{-5} \tag{270}$$

which is small enough for us to do the linearised theory.

10.1 Jeans Instability and Newtonian Cosmology

10.1.1 Fluid with Gravity

We first try to understand some of the physics just using the Newtonian gravity. We consider a non-relativistic fluid with density ρ with gravity turned on.

- **Gravity Field** the gravity potential is given by the Newton's law of gravity:

$$\Delta\varphi = 4\pi G\rho, \tag{271}$$

- **Fluid Equations** a fluid is described by a function $p = p(\rho)$ and the fluid equations are given by:

$$\begin{aligned} \frac{\partial\rho}{\partial t} + \nabla\cdot(\rho v) &= 0, \\ \frac{\partial v}{\partial t} + (v\cdot\nabla)v &= -\left(\frac{1}{\rho}\right)\nabla p - \nabla\varphi \end{aligned} \tag{272}$$

The first is the continuity equation, which describes the conservation of mass, and the second is the Euler equation, which is the equation of motion for the fluid.

One can derive the Euler equation directly from Newton's Law, the LHS is the acceleration of a fluid element, and the RHS is the force, both pressure and the gravity force.

10.1.2 Background Perturbation

Now we want to take an ansatz of linear perturbation on a homogeneous background, we write:

$$\begin{aligned} \rho &= \rho_0 + \delta\rho, \quad p = p_0 + \delta p, \\ \varphi &= \varphi_0 + \delta\varphi, \quad v = v_0 + \delta v \end{aligned} \tag{273}$$

notice that the 0 part is only time dependent which is homogeneous, and the delta part is spacetime dependent.

- Now if we assume the universe is static, then we have:

$$v_0 = 0, \quad \Rightarrow \quad \varphi_0 = 0 \tag{274}$$

Yet however, we can see this not a solution to the EoM, nevertheless, we can just do this. Now let's perturb the EoM and keep everything to the linear order, we have:

$$
\begin{aligned}
&\Delta\delta\varphi = 4\pi G\delta\rho \\
&\frac{\partial\delta\rho}{\partial t} + \rho_0\nabla\cdot\delta v = 0, \\
&\frac{\partial\delta v}{\partial t} = -\left(\frac{1}{\rho_0}\right)\nabla\delta p - \nabla\delta\varphi,
\end{aligned} \tag{275}
$$

we can use the EOS $p(\rho)$ to eliminate δp, and we have:

$$
\delta p = v_s^2\delta\rho, \quad v_s^2 = \frac{\partial p}{\partial\rho}\ |_{\rho_0} \tag{276}
$$

where v_s is the speed of sound in the fluid. Now we take a time derivative of the second equation and a gradient of the third equation, we combine them together and we have:

$$
\frac{\partial^2\delta\rho}{\partial t^2} - v_s^2\Delta\delta\rho = 4\pi G\rho_0\delta\rho \tag{277}
$$

10.1.3 Solution of the Perturbation

Now we try to find a solution to Equation 277. The standard solution is to take a Fourier transform, we write:

$$
\rho(x,t) \sim e^{ik\cdot x - i\omega t} \tag{278}
$$

Taking this ansatz into the EoM, we have:

$$
\omega^2 = v_s^2k^2 - 4\pi G\rho_0 \tag{279}
$$

Now we focus on the time dependence of the solution. If we have:

$$
k < k_J = \sqrt{\frac{4\pi G\rho_0}{v_s^2}} \tag{280}
$$

Then the solution will be growing exponentially, which is called the **Jeans instability**. The critical scale or the critical wavelength is called the **Jeans length**:

$$
\lambda_J = \frac{2\pi}{k_J} = \sqrt{\frac{\pi v_s^2}{G\rho_0}} \tag{281}
$$

10.2 Revision of General Relativity and Picture

Now we turn into a more realistic calculation using the general relativity. Here is a big picture of how we do the calculation:

- We used the Linearised gravity, but not on the Minkowski background, but on the FRW background. And work mostly in Fourier space.
- Distinguish modes inside and outside the cosmological horizon.
- Remember that the only perturbation matters when it doesn't decay with time.

Here is a general picture of the inhomogeneities in the universe:

- Before recombination, the perturbations are mostly oscillations and can be seen in the CMB picture.
- After recombination, the purturbations are mostly growing modes and can be seen in the large scale structure picture.

10.2.1 Friedmann Equation and Energy Conservation

We now review the Friedmann equation, spacial component of EFE and the energy conservation equation. We focus on the case of flat spacial curvature and no cosmological constant. The FWR metric is given by:

$$ds^2 = a^2(\eta)(-d\eta^2 + dx^i dx^i) \tag{282}$$

Here we use the conformal time η instead of the coordinate time t, which is related by $a(\eta)d\eta = dt$. And we write the derivative with respect to the conformal time as a prime, $a(\eta)'$ and $\dot{a}(\eta)$ as derivative with respect to the coordinate time.

$$H = \frac{\dot{a}}{a} = \frac{a'}{a^2} \tag{283}$$

The EoM is given by with $\Lambda = 0$and $k = 0$:

- **Friedmann Equation**:

$$\frac{a'^2}{a^4} = \frac{8\pi G}{3}\rho \tag{284}$$

- **Spatial Component of EFE**:

$$2\frac{a''}{a^3} - \frac{a'^2}{a^4} = -8\pi G p \tag{285}$$

- **Energy Conservation**:

$$\rho' + 3\frac{a'}{a}(\rho + p) = 0 \tag{286}$$

Here we review the solutions with different EOS.

- For Radiation Dominant Universe:

$$a(\eta) = \text{const} \times \eta, \quad \eta \sim t^{\frac{1}{2}} \tag{287}$$

- For Matter Dominant Universe:

$$a(\eta) = \text{const} \times \eta^2, \quad \eta \sim t^{\frac{1}{3}} \tag{288}$$

- For Dark Energy Dominant Universe (Λ dominant):

$$a(\eta) = -\frac{1}{H\eta} \sim e^{Ht}, \quad \eta \sim -e^{-Ht} \tag{289}$$

Section 2.2.4 and Section 3.2.1 have a thorough discussion of these calculations.

10.2.2 Conformal Time of Epochs

We want to calculate the conformal time at different epochs, we mainly care about recombination, matter-radiation equality and the present time. Remember the EoM of general matter is given by Equation 45:

$$H = H_0\sqrt{\Omega_{\text{rad}}\left(\frac{a_0}{a}\right)^4 + \Omega_M\left(\frac{a_0}{a}\right)^4 + \Omega_\Lambda} \tag{290}$$

or if we used the redshift z as preferred time variable, we have:

$$z = \frac{a_0}{a} - 1, \quad a = \frac{a_0}{1+z} \tag{291}$$

Then we notice that through measurement and other calculations, we have the following values for the parameters:

$$\Omega_{\mathrm{DM}} = 0.22, \quad \Omega_{\mathrm{B}} = 0.05, \quad \Omega_{\Lambda} = 0.73, \quad \Omega_{\mathrm{rad}} = 10^{-5}, \quad |\Omega_{\mathrm{curv}}| < 0.01$$
$$\Omega_M = \Omega_{\mathrm{DM}} + \Omega_{\mathrm{B}} = 0.27 \tag{292}$$

This is shown that the spacial curvature is very small and we can just assume the space is flat. We also know that the matter-radiation equality and recombination happens at

$$z_{\mathrm{eq}} = 3400, \quad z_{\mathrm{rec}} = 1100 \tag{293}$$

a reminder from Equation 157. Now we can calculate the conformal time at these epochs, we can rewrite the EoM with redshift as:

$$\eta = \int_z^{\infty} \frac{dz'}{a_0 H_0 \sqrt{\Omega_{\mathrm{rad}}(1+z')^4 + \Omega_M(1+z')^3 + \Omega_{\Lambda}}} \tag{294}$$

This integral can be calculated numerically, and we have:

$$\frac{\eta_0}{\eta_{\mathrm{rec}}} = 51, \quad \frac{\eta_0}{\eta_{\mathrm{eq}}} = 1.2 \times 10^2, \quad \frac{\eta_{\mathrm{rec}}}{\eta_{\mathrm{eq}}} = 2.4 \tag{295}$$

These are some important numbers for the conformal time at different epochs.

10.3 Perturbation of GR

10.3.1 Convention for the Metric

Now let's try to consider the perturbation of the metric on a FRS background. We now take a $(-,+,+,+)$ convention for the metric signature. Then define:

$$ds^2 = a^2(\eta)\gamma_{\mu\nu}dx^\mu dx^\nu \tag{296}$$

and here:

$$\gamma_{\mu\nu} = \eta_{\mu\nu} - h_{\mu\nu} \tag{297}$$

> **Remark**
>
> by convention, we have the minus sign in front of $h_{\mu\nu}$

Then we take care of raising and lowering indices of the linearised metric:
- For $h_{\mu\nu}$ we raise and lower with $\eta_{\mu\nu}$, which is the Minkowski metric:

$$h^\mu_\nu = \eta^{\mu\alpha}h_{\alpha\nu}, \quad h^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta}h_{\alpha\beta} \tag{298}$$

- For $\gamma_{\mu\nu}$ we define the inverse:

$$\gamma^{\mu\nu} = \eta^{\mu\nu} + h^{\mu\nu} \tag{299}$$

- For the full metric $g_{\mu\nu}$ we similarly define the inverse:

$$g^{\mu\nu} = a^{-2}\gamma^{\mu\nu} = a^{-2}(\eta^{\mu\nu} + h^{\mu\nu}) \tag{300}$$

Then we can consider the lineaized EFE, we have:

> **Theorem 10.1** — **Linearized EFE and Energy Conservation**
>
> $$\delta G_{\mu\nu} = 8\pi G \delta T_{\mu\nu} \quad \delta\nabla_\mu T^\mu_\nu = 0 \tag{301}$$

10.3.2 Gauge Fixing

To control more of the theory, we have to do a gauge fixing. We only consider small gauge transformation, which is given by:

$$x^\mu \Rightarrow x' = x^\mu + \xi^\mu \tag{302}$$

The metric transforms as:

$$\begin{aligned} g^{\mu\nu} &\Rightarrow g'^{\mu\nu} \sim g^{\mu\nu} + \nabla^\mu \xi^\nu + \nabla^\nu \xi^\mu \\ h^{\mu\nu} &\Rightarrow h'^{\mu\nu} \sim h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu + 2\frac{\partial_\lambda a}{a}\xi^\lambda \eta^{\mu\nu} \end{aligned} \tag{303}$$

YL: [There may be mistakes in above equations, yet its not important for following discussions.]
Now we have to choose a gauge to fix the 4 DoF given by ξ^μ. Now we only fix three first:

$$h_{0i} = 0 \tag{304}$$

10.3.3 Energy Momentum Tensor

Now we have to consider the energy momentum tensor, we take a perfect fluid as a model, which is good in many cases. The energy momentum tensor is given by:

$$T^\mu{}_\nu = (\rho + p)u^\mu u_\nu + p\delta^\mu_\nu \tag{305}$$

> **Note**
>
> Notice that here we use the $(-,+,+,+)$ convention. Thus, the EM tensor is like this.

Now we take the purturbation of the energy momentum tensor, we need to take the purturbation of 3 stuffs:

$$\begin{aligned} \rho &= \rho_0(\eta) + \delta\rho \\ p &= p_0(\eta) + \delta p \\ u^\mu &= \overline{u}^\mu + \delta u^\mu \end{aligned} \tag{306}$$

Notice that the 4-velocity is not independent yet is constrained by the normalization condition:

$$g_{\mu\nu}u^\mu u^\nu = -1 \tag{307}$$

We take an ansatz of the 4-velocity as:

$$\overline{u}^\mu = \left(\frac{1}{a}, 0, 0, 0\right), \quad u^\mu = \left(\frac{1}{a}\right)(1 + \delta u^0, v^i) \tag{308}$$

Thus, from the normalization condition, we have:

$$\begin{aligned} \delta u^0 = -\frac{1}{2}h_{00}, \quad u^0 = \left(\frac{1}{a}\right)\left(1 - \frac{1}{2}h_{00}\right), \quad u_0 = -a\left(1 + \frac{1}{2}h_{00}\right) \\ u^i = \left(\frac{1}{a}\right)v^i, \quad u_i = av^i = av_i \end{aligned} \tag{309}$$

> **Remark**
>
> here we define $v_i = \delta_{ij}v^j$

This result is dependent on the gauge choice of $h_{0i} = 0$. We plug this in the energy momentum tensor, we have:

$$
\begin{aligned}
\delta T^0{}_0 &= -\delta\rho \\
\delta T^0{}_i &= (\rho_0 + p_0) v_i \\
\delta T^i{}_j &= \delta p \delta_{ij}
\end{aligned}
\tag{310}
$$

The energy conservation equation is given by:

$$
\begin{aligned}
\delta\rho' + 3\frac{a'}{a}(\delta\rho + \delta p) + (\rho + p)\left(\partial_i v_i - \frac{1}{2}h'\right) = 0 \\
\partial_i \delta p + (\rho + p)\left(4\frac{a'}{a}v_i + \frac{1}{2}\partial_i h_{00}\right) + [v_i(\rho + p)]' = 0,
\end{aligned}
\tag{311}
$$

where $h = h_{ii}$ is the trace of the spatial part of the metric perturbation.

YL: [I haven't checked this equation personally, yet following we will have simpler ones, this is just for completeness.]

11 Lecture 11: Cosmological Perturbation Theory II

Though working in linear perturbation theory, the EFE and Energy conservation equations are still too complicated to solve, and we now also have a residue gauge freedom to fix. Thus, to do perturbation theory, we have to make some fuether more simplifying assumptions.

This can be done by doing mode expansions and helicity decomposition, which will be the topic of next section.

11.1 Fourier Transformation and Helicity Decomposition

11.1.1 Fourier Space

The EFE and energy conservation equations are linear, so we can do a Fourier transformation to the variables and each mode will evolve independently. Thus, we can write the perturbations as:

$$h_{\mu\nu}(\eta, x) = \int d^3k e^{ik\cdot x} h_{\mu\nu}(\eta, k) \tag{312}$$

and we do the same for the matter perturbations $\delta p, \delta\rho, v_i$. Notice that we only mode expand the spacial coordinates not the time coordinate.

The EoMs are linear thus we expect we can just take:

$$\partial_i \to ik_i \tag{313}$$

Notice that the mode may have an interpretation of momentum, but its not true, for we perturb on a curved background, thus the **physical momentum** is in fact:

$$q(\eta) = \frac{k}{a(\eta)} \tag{314}$$

11.1.2 Helicity Decomposition

Another important observation is that the equations of motion with a fixed mode can be further decomposed. The components of the original EoM are SO(3) tensors—not generally covariant ones, since we have already fixed much of the gauge freedom. Once we perform the mode expansion and pick out a definite mode, the components transform as SO(2) tensors under the residual symmetry of rotations around k

Consequently, a tensor equation can be decomposed into a set of equations labeled by definite helicity, and modes of different helicities do not mix. To simplify the EoM further, it is therefore sufficient—and much more convenient—to study a single mode of fixed helicity at a time.

Now let's see which components of the dynamical variables are helicity-0, helicity-1, and helicity-2 modes.

- **Helicity-0** (SO(2) Scalars)
  - Stuffs that are originally scalars: $h_{00}, \delta p, \delta\rho$
  - SO(3) vectors parallel to k_i
  - SO(3) tensors parallel to $k_i k_j, \delta_{ij}$
- **Helicity-1** (SO(2) Vectors)
  - SO(3) vectors perpendicular to k_i, eg. v^T satisfying $k_i v_i^T = 0$
  - SO(3) tensors that are symmetric, traceless, and transverse to k_i, eg. h_{ij}^T satisfying $h_{ii}^T = 0, k_i {h^T}^i{}_j = 0$

11.2 Scalar Perturbations

Starting this section, we will only focus on the helicity-0 modes, which are called **scalar perturbations**.

11.2.1 Scalar Components

Let's have a look at the scalar components of the perturbation.

- Decomposing the metric perturbation:

$$\begin{aligned} h_{00}(\eta, k) &= 2\phi(\eta, k) \\ h_{ij} &= -2\psi\delta_{ij} - 2k_i k_j E + i\left(k_i W_j^T + k_j W_i^T\right) + h_{ij}^{TT} \end{aligned} \tag{315}$$

Where W_i^T satisfies $k_i W_i^T = 0$ and h_{ij}^{TT} satisfies $h_{ii}^{TT} = 0, k_i h_{ij}^{TT} = 0$. We can see that the scalar perturbations are:

$$\varphi, \quad \psi, \quad E \tag{316}$$

- Decomposing the velocity perturbation:

$$v_i = v_i^T + ik_i v \tag{317}$$

where v_i^T satisfies $k_i v_i^T = 0$. We can see that the scalar perturbation is:

$$v \tag{318}$$

- Decomposing the energy-momentum tensor perturbation:

Using the perfect fluid form of the energy-momentum tensor, all two dof are scalar perturbations:

$$\delta p, \quad \delta\rho \tag{319}$$

Thus, in total, we have 6 scalar perturbations.

11.2.2 Gauge Fixing

Before we choose 3 gauge conditions, namingly $h_{0i} = 0$, now we choose the forth one to make life simpler. Now the leftover gauge transformation can be written as :

$$\xi_0 = \partial_\eta \sigma, \quad \xi_i = -\partial_i \sigma \tag{320}$$

Now we can choose an appropriate σ to set $E = 0$. With this gauge fixing choice, the metric perturbation becomes:

$$ds^2 = a^2(\eta)\left[-(1+2\phi)d\eta^2 + (1+2\psi)\delta_{ij}dx^i dx^j\right] \tag{321}$$

> **Remark**
>
> remember we take the convention of $\gamma = \eta - h$

11.2.3 EoMs

Now we can write down the EoMs for the scalar perturbations. The EFE gives us 4 equations, and the energy-momentum conservation gives us 2 equations, and the EOS gives us 1 equation, thus we have 7 equations. (remember that EFE is not independent of the energy-momentum conservation, thus we don't have too much equation.)

- Einstein Field Equations:

$$
\begin{aligned}
&\phi = -\psi \\
&-k^2\phi - 3\frac{a'}{a}\phi' - 3\frac{a'}{a^2}\phi = 4\pi G a^2 \cdot \delta\rho_{tot} \\
&\phi' + \frac{a'}{a}\phi = -4\pi G a^2 \cdot [(\rho + p)v]_{tot} \\
&\phi'' + 3\frac{a'}{a}\phi' + \left(2\frac{a''}{a} - \frac{a'^2}{a^2}\right)\phi = 4\pi G a^2 \cdot \delta p_{tot}
\end{aligned}
\tag{322}
$$

Here notice that we don't assume that the universe have only one component but many components, thus the energy density and pressure are the total ones.

$$
\delta\rho_{tot} = \sum_\lambda \delta\rho_\lambda, \quad \delta p_{tot} = \sum_\lambda \delta p_\lambda, \quad [(\rho + p)v]_{tot} = \sum_\lambda (\rho_\lambda + p_\lambda)v_\lambda \tag{323}
$$

notice that here v_λ is the scalar component of the velocity perturbation of the lambda component of matter, don't get confused with the total velocity perturbation.

- Energy-Momentum Conservation:

$$
\begin{aligned}
&\delta\rho'_\lambda + 3\frac{a'}{a}(\delta\rho_\lambda + \delta p_\lambda) + (\rho_\lambda + p_\lambda)(\Delta v_\lambda - 3\phi') = 0 \\
&[(\rho_\lambda + p_\lambda)v_\lambda]' + 4\frac{a'}{a}(\rho_\lambda + p_\lambda)v_\lambda + \delta p_\lambda + (\rho_\lambda + p_\lambda)\phi = 0,
\end{aligned}
\tag{324}
$$

Here each equation is true for each component.

- Equation of State:

$$
\delta p_\lambda = u_{s,\lambda}^2 \delta\rho_\lambda \tag{325}
$$

Notice that for any matter we always have an equation of state, generally we write the ansatz:

$$
p_\lambda = w_\lambda(\rho_\lambda)\rho_\lambda \tag{326}
$$

Remember generally w_λ is a function of ρ_λ, thus depend on time. And thus the perturbation may have the above form of equation of state, where $u_{s,\lambda}^2$ is the sound speed square of the lambda component of matter. In general, $u_{s,\lambda}^2$ is not equal to w_λ, and it can be a function of ρ_λ, thus also depend on time.

11.2.4 Subhorizon and Superhorizon Modes

The solution to above EoMs may vary depends on modes. We distinguish two kinds of modes, which give rise to distinct behaviors of the perturbations. We use two parameters to characterize the modes:

$$
q(\eta) = \frac{k}{a(\eta)}, \quad H(\eta) = \frac{a'(\eta)}{a^2(\eta)} \tag{327}
$$

If $q(\eta) \gg H(\eta)$, we call it a **subhorizon mode**, and if $q(\eta) \ll H(\eta)$, we call it a **superhorizon mode**.

- In Superhorizon modes $q \ll H$. The expansion of the universe dominates the evolution, thus the perturbations are rather small, the univse is rather homogeneous and isotropic.
- In Subhorizon modes $q \gg H$. The expansion of the universe is negligible, thus the perturbations can grow and become large, the universe is rather inhomogeneous and anisotropic.

For a fixed mode we can calculate:

$$\frac{q(\eta)}{H(\eta)} = \frac{k}{(a'(\eta)/a(\eta))} = \frac{k}{a(\eta)H(\eta)} \tag{328}$$

For a radiation or matter dominated universe, $a < c \cdot t, H \sim \frac{1}{t}$, as times goes q/H grows. Thus, at some time superhorizon modes will become subhorizon modes, and the perturbations will grow. This is called **Entry the Horizon**.

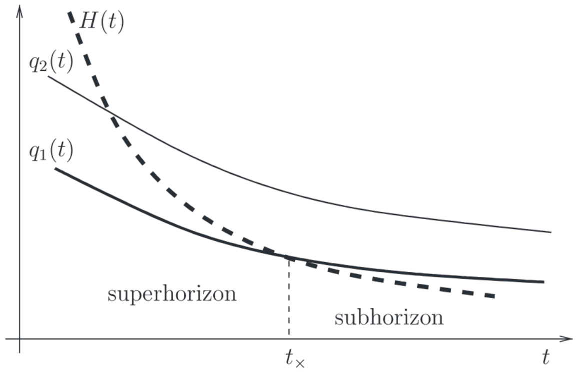

Figure 14: The evolution of a mode from superhorizon to subhorizon.

11.3 Single Component Universe

Now let's assume that the universe only have one component of matter, and try to solve the EoMs for different types of matter.

11.3.1 Equation for Metric Perturbation

We take the 2nd and 4th EFE equations, and the EOS, we try to make a closed equation with ϕ the only variable. We can get:

$$\phi'' + 3\frac{a'}{a}(1+u_s^2)\phi' + \left[2\frac{a''}{a} - \left(\frac{a'}{a}\right)^2(1-3u_s^2)\right]\phi + u_s^2k^2\phi = 0 \tag{329}$$

Notice that this is sort of a harmonic oscillator equation with:

$$\begin{gathered} 3\frac{a'}{a}(1+u_s^2)\phi' \text{ as fraction} \\ 2\frac{a''}{a} - \left(\frac{a'}{a}\right)^2(1-3u_s^2) \text{ as mass term} \end{gathered} \tag{330}$$

As discussed before, we now can see explicitly that the behavior of solution is depend on the mode, whether it is subhorizon or superhorizon.

For the case of:

$$u_s q = u_s\frac{k}{a} \ll H \tag{331}$$

The last term $u_s^2k^2$ is negligible. However, as:

$$u_s q = u_s\frac{k}{a} \gg H \tag{332}$$

we know the last term $u_s^2k^2$ dominates. This comparison can be written as one between the physical wavelength of the mode and the **sound horizon**:

$$\lambda_s \equiv \frac{u_s}{H} \tag{333}$$

We list the relation here:

- For $\lambda \gg u_s H^{-1}$ (superhorizon) then $u_s q \ll H$ thus we have the finial term negligible.
- For $\lambda \ll u_s H^{-1}$ (subhorizon) then $u_s q \gg H$ thus we have the finial term dominates.

> **Note**
>
> The sound horizon also have a physical meaning as the distance that a sound wave travel after the big bang.

11.3.2 Superhorizon Modes

Now use the Friedmann equation and the spacial EFE with $k = \Lambda = 0$,Equation 284,Equation 285, we see the mass term is related to the background matter component as:

$$2\frac{a''}{a} - \frac{a'}{(a)^2}(1 - 3u_s^2) = -8\pi G a^2 (p - u_s^2 \rho) \tag{334}$$

Now we make an simple assumption:

- For the component we have $p = \omega\rho$ and $u_s^2 = \omega$.

> **Remark**
>
> Claimed in the book that this is true for most hot big bang epoch

Thus this "mass" term vanishes, and the equation becomes:

$$\phi'' + 3\frac{a'}{a}(1 + u_s^2)\phi' + u_s^2 k^2 \phi = 0 \tag{335}$$

Now take the case that $\lambda \gg u_s H^{-1}$, the finial term is negligible, thus we have:

$$\phi'' + 3\frac{a'}{a}(1 + u_s^2)\phi' = 0 \tag{336}$$

This equation have two solutions:

$$\begin{aligned} &\phi = \text{const} \\ &\phi \sim \eta^{1-C} \text{ where } C > 1 \end{aligned} \tag{337}$$

The first is constant, and the second goes to infinity as η goes to 0, while decays as η grows. Thus, we call the second solution the **decaying mode**. We don't care about the decaying mode, thus we see that the metric perturbation is constant.

Now using the 2nd EFE and Equation 284, we can get a relation between the matter perturbation and the metric perturbation, we define the **relative perturbation** as:

$$\delta = \frac{\delta\rho}{\rho} \tag{338}$$

According to above equations, we have:

$$\delta = -2\phi \tag{339}$$

we can see its true that superhorizon modes doesn't effect the inhomogeneity of the universe, thus the universe is rather homogeneous and isotropic.

11.3.3 Subhorizon Modes I: Relativistic Matter

Now we consider the case of $\lambda \ll u_s H^{-1}$, thus the last term dominates, and we have:

$$a \sim \eta \quad \omega = u_s^2 = \frac{1}{3} \tag{340}$$

Thus the equation becomes:

$$\phi'' + \frac{4}{\eta}\phi' + \frac{1}{3}k^2\phi = 0 \tag{341}$$

This equation admit a general solution (which doesn't decay) as:

$$\phi(\eta) = -3\phi_{(i)} \cdot \frac{1}{(u_s k\eta)^2}\left[\cos(u_s k\eta) - \frac{\sin(u_s k\eta)}{u_s k\eta}\right] \tag{342}$$

If consider the limit of $u_s k\eta \gg 1$ (subhorizon), we have:

$$\phi(\eta) \sim -3\phi_{(i)} \cdot \frac{1}{(u_s k\eta)^2}\cos(u_s k\eta) \tag{343}$$

The now we consider the **relative perturbation**, as $k \to \infty$ using the second EFE, we have:

$$\begin{aligned} \delta\rho &= -\frac{1}{4\pi G}\left(\frac{k^2}{a^2}\right)\phi \\ \delta &= \frac{\delta\rho}{\rho} \sim 6\phi_{(i)} \cdot \cos(u_s k\eta) \end{aligned} \tag{344}$$

Thus we have the conclusion that :

- At radiation dominated era, there is NO jean instability, the perturbation will just oscillate and never grow.

11.3.4 Subhorizon Modes II: Non-relativistic Matter

Now we consider the case of non-relativistic matter, and we know:

$$\omega = u_s^2 = 0, \quad a \sim \eta^2 \tag{345}$$

Thus the equation becomes:

$$\phi'' + 3\frac{a'}{a}\phi' = 0 \tag{346}$$

Indeed, the solution is again a constant:

$$\phi = \text{const} \tag{347}$$

Now we consider the relative perturbation, using the 2nd EFE we can get:

$$\delta\rho = -\frac{1}{4\pi G a^2}\left(k^2 + \frac{12}{\eta^2}\right)\phi \tag{348}$$

- In **Superhorizon** limit $k\eta \ll 1$, we have:

$$\delta\rho \sim \frac{1}{a^3} \Rightarrow \delta = \frac{\delta\rho}{\rho} \sim \text{const} \tag{349}$$

- In **Subhorizon** limit $k\eta \gg 1$, we have:

$$\delta\rho \sim \frac{1}{a^2} \Rightarrow \delta \sim a \tag{350}$$

Thus we have the conclusion that:

- At matter dominated era, there is jean instability, the perturbation will grow as the universe expands. Though not exponentially but as a power law.

12 Lecture 12: CMB Anisotropies

12.1 Multi Component Universe

Now the actual universe is not just made of one component, but it is made of multiple components. At radiation dominated era, we have the perturbation of DM and Baryons.

An important point to know is that the there are two kinds of initial conditions for the modes:

- **Adiabatic mode**: initial condition for $\varphi_i \neq 0$ and other perturbations are 0.
- **Isocurvature mode**: initial condition for $\varphi_i = 0$ and other perturbations are not 0.

12.2 CMB Anisotropies

An obervation fact is that we can measure the CMB at different angles. The spectrum gives us the effective temperature of CMB. However, observation shows that the effective temperature is not isomorphic. We define:

$$\frac{\delta T(\theta, \varphi)}{T_0} \tag{351}$$

The strongest anisotropy can be explained by the moving of earth, however, there are still some anisotropies, which is contributed by the intrinsic anisotropy of the CMB. If we consider the anisotropy of CMB temperature in the forier space of spherical harmonics, we can find the following kinds of relation:

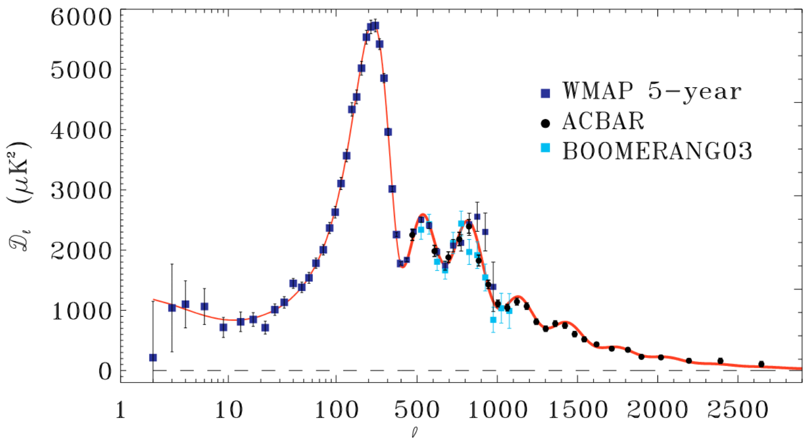

Figure 15: Anisotropy of CMB temperature.

We try to understand how this is the case.

12.2.1 Estimate the First Peak

We can assume that this kind of anisotropy is given by the occilation of subhorizon modes of relativitic matter at relativistic dominated era Section 11.3.3. The occilation pattern is given by the following equation:

$$\delta_{\text{rad}} \sim 6\varphi_{(i)} \cos(u_s k\eta) \tag{352}$$

and at recombination, $\eta = \eta_r$, the light freeze out and propagate freely to us, as shown in the following figure.

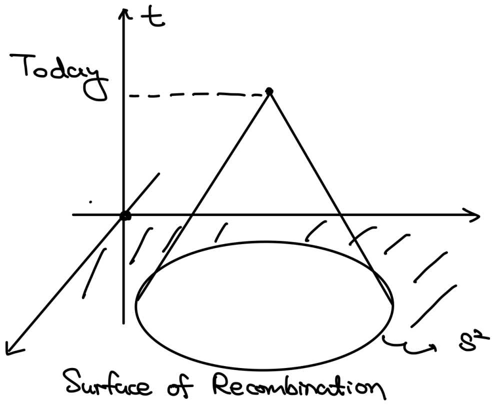

Figure 16: Propagation of CMB photons.

If we assume that the temperature anisotropy is strongly contributed by the occilation of subhorizon modes, say:

$$\frac{\delta T}{T_0} \sim \delta_{\mathrm{rad}} \sim 6\varphi_{(i)} \cos(u_s k \eta_r) \tag{353}$$

Then we may understand the occilation of Figure 15 is given by the cosine function. This is indeed the case and we can estimate the position of the first peak.

The mode that contributes to the first anisotropy peak at the recombination time should be:

$$u_s k \eta_r = \pi \tag{354}$$

remember at the radiation dominated era, $u_s = \frac{1}{\sqrt{3}}$, so we have:

$$k = \sqrt{3}\frac{\pi}{\eta_r} \tag{355}$$

Then we can estimate the typical angle of this mode by the relation:

$$\Delta x = \frac{2\pi}{k} = \frac{2}{\sqrt{3}}\eta_r \quad \Delta x = \Delta\theta \times \eta_0 \tag{356}$$

The second relation is estimated from the following figure:

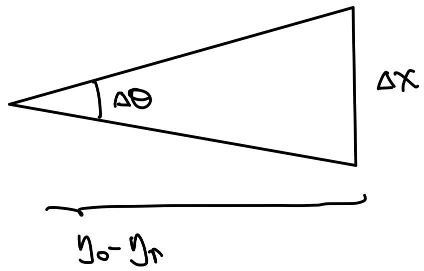

Figure 17: The relation between the physical distance and the angle.

and $\eta_r \ll \eta_0$, thus we eventually have:

$$\Delta\theta = \frac{2}{\sqrt{3}}\left(\frac{\eta_r}{\eta_0}\right) \tag{357}$$

Then we estimate the angular fourier mode of this typical angle:

$$l = \frac{2\pi}{\Delta\theta} \sim 300 \tag{358}$$

This is of the same order of the position of the first peak in Figure 15, which is around $l = 200$.

12.2.2 Gaussian Random

Let's now treat the CMB temperature anisotropy more carefully. In modern cosmology, we believe that the initial perturbation $\varphi_{(i)}$ of the energy density δ is picked from some random distributions (that's why the CMB looks so random). Thus, if we think of CMB temperature as a result of perturbation, it then should depend on the initial perturbation $\varphi_{(i)}$, thus should be in some random distribution.

We can make a fourier transformation of the CMB temperature anisotropy:

$$\frac{\delta T(\mathbf{n})}{T} = \sum_{l=1}^{\infty} \sum_{m=-l}^{m=l} a_{lm} Y_{lm}(\mathbf{n}). \tag{359}$$

The coefficients should be picked from some random distribution, in this discussion. We now assume:

- The distribution is isotropic and the temperature is just picking out numbers from the distribution.

The isotropy of the distribution tells us that:

$$\langle a_{lm} a^*_{l'm'} \rangle = C_l \delta_{ll'} \delta_{mm'} \tag{360}$$

where we average over the distribution. In fact, observation shows that the distribution is a Gaussian. Thus, the number C_l completely determines the distribution and all non-random information of the CMB temperature.

The fact that we can measure a distribution from a random picked result is strange. However, though we only have one universe, different m of CMB temperature can be thought as different picks form the distribution. Thus, in fact we can get the quantity C_l by averaging over m:

$$C_l = \left(\frac{1}{2l+1}\right) \sum_{m=-l}^{m=l} a_{lm} a^*_{lm} \tag{361}$$

The above figure Figure 15 is just the plot of l with a quantity proportional to C_l.

13 Lecture 13: Basics of Inflation

13.1 Motivation: Horizon Problem

Remember in the section before, we have noticed that the initial perturbation of the universe metric is a random number from a Gaussian distribution. Say:

$$\delta_{\text{rad}}(k) = 6\varphi_{(i)}(k)\cos(u_s k\eta) \tag{362}$$

and it satisfy the distribution:

$$P\Big(\varphi_{(i)}(k)\Big) \sim \exp\left(-\frac{\varphi_{(i)}(k)^2}{2\sigma(k)^2}\right) \tag{363}$$

where $\sigma(k)$ is the variance of the distribution that due to the isotropic assumption only depends on the magnitude of k. This distribution leads to the following correlation of the initial perturbation:

$$\langle\varphi_{(i)}(k)\varphi^*_{(i)}(k')\rangle \sim \sigma(k)^2\delta^3(k-k') \tag{364}$$

Also, with this ensemble we can estimate the two point correlation of the radiation perturbation:

$$\langle\delta_{\text{rad}}(k)\delta_{\text{rad}}(k')\rangle \sim \cos^2(u_s k\eta)\sigma(k)^2\delta^3(k+k') \tag{365}$$

> **Note**
>
> Note that the delta function is $\delta^3(k+k')$ is due to the fact that the perturbation is real. If we want only the norm of the mode to be correlated, this directly leads to the delta function of $\delta(k)\delta^*(k)$ which is $\delta(k)\delta(-k)$.

In fact, observations shows that the correlation of the perturbation (at the last scattering surface) is given by a power law:

$$\langle\delta_{\text{rad}}(k)\delta_{\text{rad}}(k')\rangle \sim \delta^3(k+k')10^{-10}\left(\frac{k_*}{k}\right)^{3+(n_s-1)} \tag{366}$$

However, in fact this correlation causes a problem. If we read this observation result we know that:

1. The perturbation is small (10^{-5}) and thus the universe is very homogeneous at the beginning.
2. The perturbation has simple power spectrum.
3. The perturbation is correlated at small k, which means that the perturbation is correlated at large scale.

The last point is a conflict with the standard cosmology we have discussed. Some long range point at the last scattering surface (LSS) will have past light cones that do not intersect, thus they cannot be correlated. However, observation tells us they are, due to the structural behavior at small k. This is called the **horizon problem**.

YL: [I still don't understand what is the FRW penrose diagram look like, why it doesn't meet????]

To resolve this problem, we need to find a mechanism that can generate correlation at large scale. This is the motivation of inflation. Inflation is a theory for period before the big bang, which tells us why we have the power spectrum P. There are several features of inflation:

- It is a period of quasi-exponential expansion of the universe

The homogeneous expansion can be given as a single field slow roll inflation mode. And the perturbation can be given by the quantum fluctuation of the inflaton field.

13.2 Inflation Idea

13.2.1 dS and Horizon Problem

We assume that the universe is dS before the big bang. Remember in dS spacetime:

$$a(t) = e^{Ht}, \quad H = \sqrt{\frac{\Lambda}{3}} = \text{constant} \tag{367}$$

In this case, we remember that we define superhorizon and subhorizon by comparing:

$$q(\eta) = \frac{k}{a(\eta)}, \quad H(\eta) = \frac{a'(\eta)}{a^2(\eta)} \tag{368}$$

Now in a dS spacetime, we have:

$$H = \text{const} \quad a(\eta) = -\frac{1}{H\eta} \tag{369}$$

Thus the modes are initially subhorizon, as the universe expands, the modes becomes superhorizon. This means that the modes can be correlated at large scale, because they are initially subhorizon and thus can interact with each other. This is how dS can solve the horizon problem.

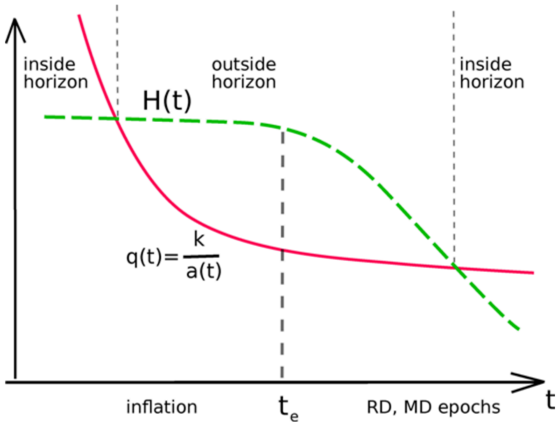

Figure 18: Illustration of solving Horizon Problem with dS.

13.2.2 Slow Roll Inflation

A problem of dS is that if it is once dominant then it will always be dominant, we can see it from the EoM of Hubble parameter:

$$H = H_0\sqrt{\Omega_\Lambda + \Omega_m(1+z)^3 + \Omega_{\text{rad}}(1+z)^4}. \tag{370}$$

as the universe expands z goes to zero. If Ω_Λ dominates at some point, it will always dominate. Thus, we need a mechanism to end the dS phase, which is called **slow roll inflation**.

Here we introduce an approach called single field slow roll inflation. We introduce a scalar field called the inflaton field, which has a potential $V(\varphi)$ with the following properties:

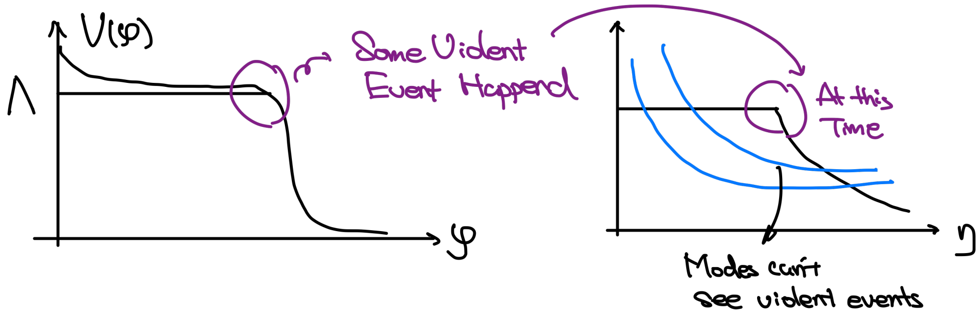

Figure 19: The potential of the inflaton field.

Initially, it is placed on the part with higher energy, which gives us the dark energy as the Cosmological constant. Then it rolls down the potential, and the cosmological constant get smaller yet the energy heat up the universe and make it radiation dominated. This is called the reheating process. After the reheating process, the universe is in the standard cosmology phase.

Unfortunately, modes of fluctuation doesn't see the reheating process, they are in super-horizon during the process, thus they are not affected by the reheating process.

Now we study the dynamics of the inflaton field. It can be described as a scalar field in dS spacetime, with the following action:

$$S = \int d^4x\sqrt{-g}\left[\frac{1}{2}g^{\mu\nu}\partial_\mu\varphi\partial_\nu\varphi - V(\varphi)\right]. \tag{371}$$

Here the metric we use is the flat slicing dS metric, which is given by:

$$ds^2 = -dt^2 + e^{2Ht}dx^2 \tag{372}$$

- Equation of Motion of the inflaton field

We can calculate the EoM of the inflaton field:

$$\ddot{\varphi} + 3H\dot{\varphi} + V'(\varphi) = 0 \tag{373}$$

Notice that we assume that the field is homogeneous, thus the spatial derivative term vanishes.

- Einstein Equation

The energy momentum tensor of the inflaton field is:

$$T_{\mu\nu} = \partial_\mu\varphi\partial_\nu\varphi - g_{\mu\nu}\mathcal{L} \tag{374}$$

Then we can calculate the energy density and pressure, in cosmology we assume the inflation field is homogeneous, thus term $\Delta\varphi$ vanishes, and we have:

$$\rho = \frac{1}{2}\dot{\varphi}^2 + V(\varphi), \quad p = \frac{1}{2}\dot{\varphi}^2 - V(\varphi) \tag{375}$$

Notice that in a slow role approximation, the potential energy term will dominate and thus we have $p \approx -\rho$, which is the same as the cosmological constant. This is how slow roll inflation can give us a dS phase. Indeed the Friedmann equation is given by:

$$H^2 = \frac{8\pi G}{3}\left[\frac{1}{2}\dot{\varphi}^2 + V(\varphi)\right] = \frac{8\pi}{3M_{\text{pl}}^2}\left[\frac{1}{2}\dot{\varphi}^2 + V(\varphi)\right] \tag{376}$$

in the inflation theory we commonly use M_{pl} instead of G, which is defined as:

$$M_{\text{pl}} = \sqrt{\frac{\hbar c}{G}} = \frac{1}{\sqrt{G}} \tag{377}$$

Now we can discuss the slow roll approximation. We need two conditions for the approximation to be valid:

- **Fraction is Large**, to have the field rolling down the potential slowly, we need the friction term $3H\dot{\varphi}$ to be large, this gives us:

$$\left|\frac{\ddot{\varphi}}{3H\dot{\varphi}}\right| \ll 1 \tag{378}$$

- **Potential Dominates**, the initial potential energy should be large enough to make the universe in a dS phase, thus we need:

$$\frac{\dot{\varphi}^2}{2V(\varphi)} \ll 1 \tag{379}$$

Under these two conditions, we can see that the EoM of inflation field will be dominated by the friction and potential term, thus we have:

$$\begin{aligned} \dot{\varphi} &= -\frac{1}{3H}V'(\varphi) \quad \text{Equation of Motion (SR)} \\ H &= \frac{1}{M_{\text{pl}}}\left(\frac{8\pi V}{3}\right)^{1/2} \quad \text{Friedmann Equation (SR)} \end{aligned} \tag{380}$$

14 Lecture 14: Inflation and Quantum Fluctuation

14.1 Inflation at Background Level

14.1.1 Slow Roll Parameters

Remember in the slow roll inflation, we have two slow roll conditions:

$$\left|\frac{\ddot{\varphi}}{3H\dot{\varphi}}\right| \ll 1 \quad \frac{\dot{\varphi}^2}{2V(\varphi)} \ll 1 \tag{381}$$

And we have two slow approximation EoMs:

$$\begin{aligned} \dot{\varphi} &= -\frac{1}{3H}V'(\varphi) \quad \text{Equation of Motion (SR)} \\ H &= \frac{1}{M_{\rm pl}}\left(\frac{8\pi V}{3}\right)^{1/2} \quad \text{Friedmann Equation (SR)} \end{aligned} \tag{382}$$

We want to express the slow roll condition purely in terms of the potential, thus we see using the EoM we can get a relation between the kinetic energy and potential energy:

$$\dot{\varphi} = -\frac{M_{\rm pl}}{(24\pi)^{1/2}}\frac{V'}{V^{1/2}}. \tag{383}$$

Taking it into the second slow roll condition, we can get the first slow roll parameter:

$$\varepsilon \equiv \frac{M_{\rm pl}^2}{48\pi}\left(\frac{V'}{V}\right)^2 \ll 1 \tag{384}$$

Then we take a derivative of Equation 383 with respect to time, we can get the relation:

$$\ddot{\varphi} = -\frac{M_{\rm pl}}{(24\pi)^{1/2}}\left(\frac{V''}{V^{1/2}} - \frac{1}{2}\frac{V'^2}{V^{3/2}}\right)\dot{\varphi} = -\frac{M_{Pl}^2}{V'^{3/2}}\left(\frac{V''}{V} - \frac{1}{2}\left(\frac{V'}{V}\right)^2\right)H\dot{\varphi}, \tag{385}$$

The first slow roll condition can be expressed as:

$$\eta \equiv \frac{M_{\rm pl}^2}{8\pi}\left(\frac{V''}{V}\right) \ll 1 \tag{386}$$

We can see that if a potential have small slow roll parameters, then it can drive a period of inflation.

14.1.2 Examples of Inflationary Potentials

For example, in fact the quadratic potential

$$V(\varphi) = \frac{1}{2}m^2\varphi^2 \tag{387}$$

can drive inflation, and the slow roll conditions tells us:

$$\frac{M_{\rm pl}^2}{\varphi^2} \ll 1 \tag{388}$$

Moreover, the energy cannot be too big, otherwise the quantum gravity effect will be important, thus we have:

$$m^2\varphi^2 \ll M_{\rm pl}^4 \tag{389}$$

Thus, we need field configuration to be extremely large, yet the mass of field should be very small.

14.2 Inflation Perturbations

Now we focus on the perturbation during inflation.

14.2.1 Massless Scalar Field in Minkowski Spacetime

The Lagrangian is given by:

$$S_\varphi = \frac{1}{2}\int d^4x\eta^{\mu\nu}\partial_\mu\varphi\partial_\nu\varphi = \frac{1}{2}\int d^4x\left[(\partial_t\varphi)^2 - (\partial_i\varphi)^2\right], \tag{390}$$

And the Hamiltonian is given by:

$$E = \frac{1}{2}\int d^3x\left[(\partial_t\varphi)^2 + (\partial_i\varphi)^2\right]. \tag{391}$$

The EoM leads to a mode expansion of the field:

$$\varphi(x,t) = \int \frac{d^3q}{(2\pi)^{3/2}\sqrt{2\omega_q}}\left(e^{i\omega_q t - iqx}A_q^\dagger + e^{-i\omega_q t + iqx}A_q\right) \tag{392}$$

And now q, x are all three dimensional vectors and $\omega_q = |q|$. We do QFT by promoting A_q and $A_q^\dagger$ to operators, and we have the commutation relation:

$$\left[A_q, A_{q'}^\dagger\right] = \delta^3(q - q') \tag{393}$$

With the on shell mode expansion, the Hamiltonian can be expressed as:

$$E = \int d^3q\omega_q\left(A_q^\dagger A_q\right) \tag{394}$$

Now we are interested in the vaccum equal time correlation function of the field:

$$\langle 0|\varphi(x,t)\varphi(y,t)|0\rangle \tag{395}$$

By the on shell mode expansion, we can get:

$$\langle 0|\varphi(x,t=0)^2|0\rangle = \int \frac{d^3q}{(2\pi)^3 2\omega_q} = \int_0^\infty \frac{q^2}{(2\pi)^2}\frac{dq}{q} \tag{396}$$

14.2.2 Inflation Perturbations

Now we view the inflation field as following;

$$\varphi_{\text{full}}(x,t) = \phi_c(t) + \varphi(x,t). \tag{397}$$

where ϕ_c is the classical homogeneous background. We consider the perturbation dynamical yet the background solution is fixed, and we consider the effective action of the perturbation, which is given by:

$$\begin{aligned} S_\varphi &= \frac{1}{2}\int d^4x\sqrt{-g}\left[g^{\mu\nu}\partial_\mu\varphi\partial_\nu\varphi - V''(\phi_c)\varphi^2\right] \\ &= \frac{1}{2}\int dtd^3xa^3\left[\dot\varphi^2 - a^{-2}(\partial_i\phi)^2 - V''(\phi_c)\varphi^2\right]. \end{aligned} \tag{398}$$

In fact in the slow roll approximation, we have V'' small that we can ignore it at the leading order. Thus, the EoM of the perturbation is given by:

$$\ddot{\varphi} + 3H\dot{\varphi} - \frac{1}{a^2}\partial_i\partial_i\varphi = 0 \tag{399}$$

We often work in conformal time, thus we can write the EoM as:

$$\varphi'' + 2\frac{a'}{a}\varphi' - \Delta\varphi = 0, \quad \text{where } \Delta = \partial_i\partial_i \tag{400}$$

We now expand the perturbation φ in terms of Fourier modes, remember that in dS $a(\eta) \sim -\frac{1}{H\eta}$. Thus:

$$\Delta\varphi \sim k^2 \quad \frac{a'}{a}\varphi \sim \frac{1}{|\eta|}k \tag{401}$$

> **Remark**
>
> YL: [I still don't understand how the second relation comes out. why $\varphi' \sim k$]

Thus, we can consider two limit of the EoM:

- If $\frac{k}{|\eta|} \gg k^2$ which is $|\eta k| \ll 1$. The mode is **Outside the Horizon**. In this time, the EoM is given by:

$$\varphi'' + 2\frac{a'}{a}\varphi' = 0 \tag{402}$$

where one classical solution is a constant.

- If $\frac{k}{|\eta|} \ll k^2$ which is $|\eta k| \gg 1$. The mode is **Inside the Horizon**. The EoM is given by:

$$\varphi'' - \Delta\varphi = 0 \tag{403}$$

This means that we have a wave solution.

As time goes, we notice that for dS spacetime:

$$t \to \infty \quad \eta \to 0 \quad |\eta k| \to 0 \tag{404}$$

Thus, we see that as time goes on $|\eta|$ decreases and modes will exit the horizon, which is different from the usual radiation dominate FRW case.

Thus, we expect the perturbation to be initially a wave solution inside the horizon, then it will exit the horizon and become a constant solution.

14.2.3 Quantization of Inflation Perturbation

To further investigate, it is useful to change a variable:

$$\chi = a(\eta)\varphi \tag{405}$$

By doing some integral by parts and calculate we can get the action of χ:

$$\begin{aligned} S_\chi &= \frac{1}{2}\int d^3x d\eta \left[\left(\chi' - \frac{a'}{a}\chi\right)^2 - (\partial_i\chi)^2\right] \\ &= \frac{1}{2}\int d^3x d\eta \left[\chi'^2 - (\partial_i\chi)^2 + \frac{a''}{a}\chi^2\right], \end{aligned} \tag{406}$$

We can do an on shell mode expansion. Note that this action behaves like a time dependent ω harmonic oscillators.

- **Inside Horizon Limit** $|k\eta| \gg 1$: At $\eta \to -\infty$ limit, we can negelect the gravitational effect $\frac{a''}{a}\chi^2$ term and it just gives us the mode expansion of a free scalar field:

$$
\chi(x, \eta) = \int \frac{d^3k}{(2\pi)^{3/2}\sqrt{2k}} \left(e^{ik\eta - ikx} A_k^\dagger + e^{-ik\eta + ikx} A_k \right), \tag{407}
$$

- **Exit Horizon** $|k\eta| \sim 1$: If we consider modes that are going to exit horizon, or at time to exit the horizon, then we need

$$
\chi(x, \eta)_{\{k\}} = \int_{\{k\}} \frac{d^3k}{(2\pi)^{3/2}\sqrt{2k}} \left(e^{-ikx} \chi_k^{(+)}(\eta) A_k^\dagger + e^{ikx} \chi_k^{(-)}(\eta) A_k \right), \tag{408}
$$

in the flat spacetime limit we have $\chi_k \sim e^{\pm i\omega\eta}$. In fact we can solve this from EoM, the momentum space EoM is given by:

$$
\chi'' - \frac{2}{\eta^2}\chi + k^2\chi = 0 \tag{409}
$$

Fixing the boundary condition to match the flat spacetime limit, we can get the solution:

$$
\chi_k^{(\pm)} = e^{\pm ik\eta} \left(1 \pm \frac{i}{k\eta} \right) \tag{410}
$$

- **Outside Horizon** $|k\eta| \ll 1$: If we consider modes that are outside the horizon, then the oscillation term is negligible, and we have:

$$
\chi(x, \eta)_{\{k\}} = \int_{\{k\}} \frac{d^3k}{(2\pi)^{\frac{3}{2}}\sqrt{2k}} \left(-\frac{1}{k\eta} \right) \left(e^{-ikx + i\alpha_k} A_k^\dagger + e^{ikx - i\alpha_k} A_k \right) \tag{411}
$$

Thus we can also find that:

$$
\varphi(x, \eta)_{\{k\}} = \int_{\{k\}} \frac{d^3k}{(2\pi)^{\frac{3}{2}}\sqrt{2k}} \frac{H}{k} \left(e^{-ikx + i\alpha_k} A_k^\dagger + e^{ikx - i\alpha_k} A_k \right) \tag{412}
$$

The result is rational, since the field fluctuation is time independent outside the horizon, as we expected.

> **Remark**
>
> Notice that all these discussion is based on a certain range of modes. We write the expansion as $\chi_{\{k\}}$ shows that this expansion is only valid for some certain modes.

We now focus on the super horizon modes, and we want to know the fluctuation of the field, which is given by the equal time correlation function, the above on shell mode expansion gives us:

$$
\langle \varphi(x, \eta)^2 \rangle = \int \frac{d^3k}{(2\pi)^3 2} \left(\frac{H^2}{k^3} \right) \tag{413}
$$

In momentum space, we have:

$$
\langle \varphi(k, \eta) \varphi(k', \eta) \rangle \sim \delta^3(k + k') \left(\frac{H^2}{2k^3} \right) \tag{414}
$$

This is gives us the famous $\frac{1}{k^3}$ dependent power spectrum instead of the usual $\frac{1}{k}$ dependent in the flat spacetime.

14.2.4 From Quantum To Classical

It is important to address a picture. We usually evaluate a harmonic oscillator state whether it is in the ground state or semiclassical state by looking at the two point equal time correlation function.

Then we may notice that, if we consider the χ field:

- At outside horizon limit:

$$\langle \chi(k,\eta)\chi(k',\eta)\rangle \sim \delta^3(k+k')\frac{1}{|k\eta|^2}\frac{1}{k} \tag{415}$$

- At inside horizon limit:

$$\langle \chi(k,\eta)\chi(k',\eta)\rangle \sim \delta^3(k+k')\frac{1}{k} \tag{416}$$

At outside horizon limit the two point correlation function is much larger than the inside horizon limit:

$$\langle \chi(k,\eta)\chi(k',\eta)\rangle_{\{|k\eta|\ll 1\}} \gg \langle \chi(k,\eta)\chi(k',\eta)\rangle_{\{|k\eta|\gg 1\}} \tag{417}$$

We can interpret this as the following result that the state we are averaging upon is becoming more and more classical as time goes on, this is effected by the time dependent perturbation term in the action.

14.3 Inflation Perturbation to CMB Anisotropy

14.3.1 Metric Perturbation from Inflation Perturbation

Finally, we want to explain the energy density perturbation, which lead to the anisotropy that we observe in CMB. We develop the following argument.

If we assume that the inflation ends (reheating) when the field value reaches some critical value $\varphi_{\text{critical}}$,

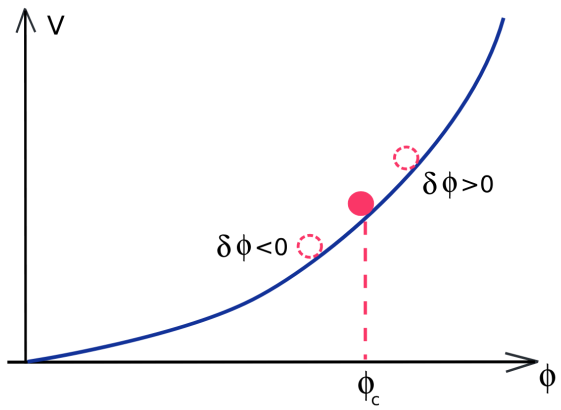

Figure 20: The end of inflation

then the presence of the perturbation will lead to that different region of the universe will end inflation at different time. Remember that we split the inflation field into a classical background and a perturbation, thus we have:

$$\varphi_{\text{full}}(x,t) = \phi_c(t) + \varphi(x,t) \tag{418}$$

for a certain x, the lag or lead of time to reach the critical value is given by:

$$\delta t_r(x) = \frac{\varphi(x,t)}{\dot{\phi}_c} \tag{419}$$

Because of this lag or lead in reheating, this cause the primordial inhomogeneity in energy density and the gravitational field. We can esitimate them from the lag or lead of reheating time, and we have:

$$\frac{\delta a}{a} = \frac{\dot{a}\delta t_r}{a} = H\delta t_r \tag{420}$$

if we plug in the dS spacetime $a(t) \sim e^{Ht}$, we can get:

$$\frac{\delta a}{a} = H\delta t_r = H\frac{\varphi(x,t)}{\dot{\phi}_c} \tag{421}$$

Note the $\dot{\phi}_c$ is related to a small parameter (slow roll parameter) ε as:

$$\dot{\phi}_c \sim \sqrt{V\varepsilon} \tag{422}$$

and from the Friedmann equation we have $HM_{\mathrm{pl}} \sim \sqrt{V}$, thus we can get:

$$\phi_{(i)} \sim \frac{\delta a}{a} \sim H\frac{\varphi(x,t)}{\dot{\phi}_c} \sim \frac{H}{\sqrt{V\varepsilon}}\varphi(x,t) \sim \frac{1}{\sqrt{\varepsilon}}\frac{\varphi(x,t)}{M_{\mathrm{pl}}} \tag{423}$$

Thus the metric perturbation is strongly related to the inflation field perturbation, given that:

$$\langle \phi_{(i)}\phi_{(i)} \rangle \sim \frac{1}{\varepsilon M_{\mathrm{pl}}^2}\langle \varphi\varphi \rangle \sim \delta^3(k+k')\frac{1}{\varepsilon M_{\mathrm{pl}}^2}\frac{H^2}{k^3} \tag{424}$$

This eventually lead to the anisotropy in we measure as Equation 366. Notice that if we want to match the observation, we need:

$$\frac{H^2}{\varepsilon M_{\mathrm{pl}}^2} \sim 10^{-10} \tag{425}$$

14.3.2 Slow roll Parameter Restriction

In reality, the correlation function is not exactly $\frac{1}{k^3}$ but with a small deviation:

$$P(k) \sim \frac{1}{k^{3+1-n_s}} \tag{426}$$

This in fact will give us a restriction on the slow roll parameter ε, η. The result is:

$$n_s - 1 = -6\varepsilon + 2\eta \tag{427}$$

Bibliography

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